University of Oslo

Summer Project – Nuclear propulsion at scale

Fuel and infrastructure requirements for an 800 ship fleet, 2035–2060 ramp

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Abstract

This study investigates the fuel and infrastructure requirements for supporting an 800 vessel nuclear shipping fleet, deployed between 2035 and 2060. This is done through a coupled maritime, reactor and supply chain model. In particular, the required natural uranium, HALEU and TRISO production and separative work are estimated, along with estimates for storage of fresh and spent fuel in ports, and transport of spent fuel out of the ports. The ship deployment follows a sigmoid curve, with a delay for the deployment of passenger-carrying ships and large fishing vessels. Each ship segment has an associated number of SMRs installed, which is related to fuel consumption through reactor burnup, operational time and refueling interval. The Allseas SMR concept is used as a basis for the modeled reactors. A simplified reactor model is employed, which defines a reactor by its power, burnup and fuel enrichment. Possible shipping routes are explored, along with a port network for service of the nuclear propelled ships. The required global fuel storage and transportation capacity is distributed between these nuclear service ports. The relationship between the fuel requirements and the nuclear supply chain is discussed.

Production of HALEU and TRISO are identified as significant bottlenecks, requiring and estimated 240 tons of HALEU per year by 2060, corresponding to 30 million TRISO pebbles. This exceeds current global production by more than an order of magnitude. Major constraints in terms of port infrastructure are identified as the storage and transportation of fuel. The largest nuclear service ports in the model would require a combined $\sim 35\ 000\ \mathrm{m}^3$ of storage capacity for fresh and spent fuel. Furthermore, these ports would need to accommodate $\gtrsim 400$ flatbed trucks annually to manage spent fuel transportation.

Moreover, a sensitivity analysis is performed to illustrate this model's sensitivity to the uncertainties of input parameters. Specifically, the burnup, fuel enrichment, refueling interval and spent fuel storage time input parameters were varied. Since the fuel enrichment and burnup for the Allseas SMR are unknown at the time of writing, the baseline assumption for these values were chosen as a middle ground between the HTR-PM and Xe-100 reactors. If the reactors installed in the fleet can achieve similar fuel burnup as the Xe-100, the fuel, storage and transportation requirements would be substantially reduced, almost by a factor 2.

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1 Introduction

International shipping is under increasing pressure to reduce greenhouse-gas emissions while maintaining reliable long-distance transport. The DNV *Maritime Forecast to 2050* describes shipping as being in the middle of a fuel and technology transition driven by regulation, commercial pressure and the need to reduce well-to-wake greenhouse-gas intensity [1].

Most decarbonisation pathways for shipping require a strong coupling between onboard technology and shoreside infrastructure. Alternative fuels such as methanol, ammonia, hydrogen or synthetic fuels may reduce emissions, but they also require large-scale fuel production, bunkering infrastructure, storage systems, safety procedures and international coordination. DNV therefore emphasises that the maritime energy transition depends on synchronised progress across ships, fuel supply and port infrastructure [1]. This infrastructure dimension is especially important for deep-sea vessels, where high energy demand, long routes and limited refuelling opportunities make fuel availability and storage capacity central constraints.

Small Modular Reactors (SMRs) have been proposed as one possible long-term option for large maritime vessels. The Allseas SMR impact study presents compact high-temperature gas-cooled reactors as a potential zero-emission power source for maritime applications, with high energy density and multi-year operation without refuelling [2]. However, nuclear propulsion does not remove the need for infrastructure. Instead, it changes the nature of the infrastructure problem: conventional bunkering is replaced by a nuclear fuel cycle involving reactor manufacturing, fuel enrichment, fuel fabrication, fuel transport, refuelling operations, spent-fuel storage and waste management. For this reason, the deployment of a nuclear-powered maritime fleet cannot be assessed only by estimating onboard power demand. It must also be connected to the supporting fuel-cycle and port-service network. A fleet that appears technically feasible from a propulsion perspective may still be limited. This project is motivated by the following systems-level question: if a large fleet of nuclear-powered vessels were deployed between 2035 and 2060, what would the corresponding maritime, reactor, fuel-cycle and port-infrastructure requirements look like?

This study develops a simplified scenario model for the deployment of an 800-vessel nuclear-powered maritime fleet between 2035 and 2060. The model is designed as a first-order infrastructure assessment: starting from a hypothetical fleet deployment pathway, it estimates the associated power demand, fuel demand, route-network expansion and port-service requirements. The study remains deliberately simplified. It does not perform reactor neutronics calculations, detailed ship-design integration, port-engineering design, transport-package licensing, safety analysis or cost optimisation. Nevertheless, the model estimates infrastructure requirements and which parts of the future nuclear maritime system may become bottlenecks.

The report is structured as follows. Section 2 introduces the main model inputs and assumptions. It describes the maritime module, the representative ship segments, the fleet-composition assumptions, operating hours, auxiliary-power assumptions and final ship-segment parameters. It also introduces the reactor assumptions used to connect ship power demand with fuel demand. Section 3 describes the supply-chain assumptions. It introduces the fuel-cycle stages considered in the model, including uranium mining and milling, conversion, HALEU enrichment and TRISO fuel fabrication. It also identifies the countries and regions treated as potential suppliers of enrichment and fuel-fabrication capacity. Section 4 presents the country and port profiles used to build the maritime network. It explains the selection of candidate ports, the geographical logic of the port network and the progressive activation of ports over time. Section 5 then describes the fleet deployment and route-network deployment, including the S-curve fleet growth, the allocation of routes by ship segment and the expansion of the maritime network between 2035 and 2060. Section 6 presents the model outputs and infrastructure calculations. It connects the fleet deployment to reactor demand, fuel throughput, enrichment requirements, refuelling demand, fresh-fuel storage, spent-fuel storage and port-based service-hub capacity. This section also introduces the assumptions used to define nuclear-capable ports and to distribute global fuel-cycle requirements across them. Section 7 presents the preliminary results. The global results describe the fleet-wide infrastructure demand, while the localized results translate those global requirements into port-level refuelling, storage and spent-fuel transport burdens. Finally, the appendix provides additional route maps for the mature 2060 network, separated by ship segment.

All model code, input files and generated outputs used in this study are stored in a dedicated GitHub repository[1].

2 Model structure and inputs

The model is organised as a scenario-based infrastructure workflow rather than as a detailed ship-design or reactor-design simulator. Starting from a set of inputs and scenario assumptions, the model first defines the nuclear-powered fleet deployment and constructs the corresponding maritime operating network. This maritime

[1] `https://github.com/mathiassig/UiO_Summer_Project--Nuclear_Propulsion`

layer determines how many ships of each segment are active in each year, how much annual sailing distance and port-call demand they generate, and which routes and ports are used by the fleet.

The resulting fleet activity is then passed to the reactor and fuel-demand layer. This layer converts ship power requirements and deployment timing into order-of-magnitude estimates of HALEU enrichment, TRISO fuel fabrication, fresh-fuel storage, spent-fuel storage and spent-fuel transport requirements.

Finally, the model connects the global fuel-cycle requirements back to the maritime network through a nuclear port-service hub layer. A subset of operating ports is assumed to become nuclear-capable over time. These ports are assigned activation years, relative service weights and maturity factors, and the global refuelling, storage and transport requirements are distributed among the active nuclear-capable ports. Figure 1 summarises this workflow and shows how the maritime, reactor, fuel-cycle and port-service components are connected.

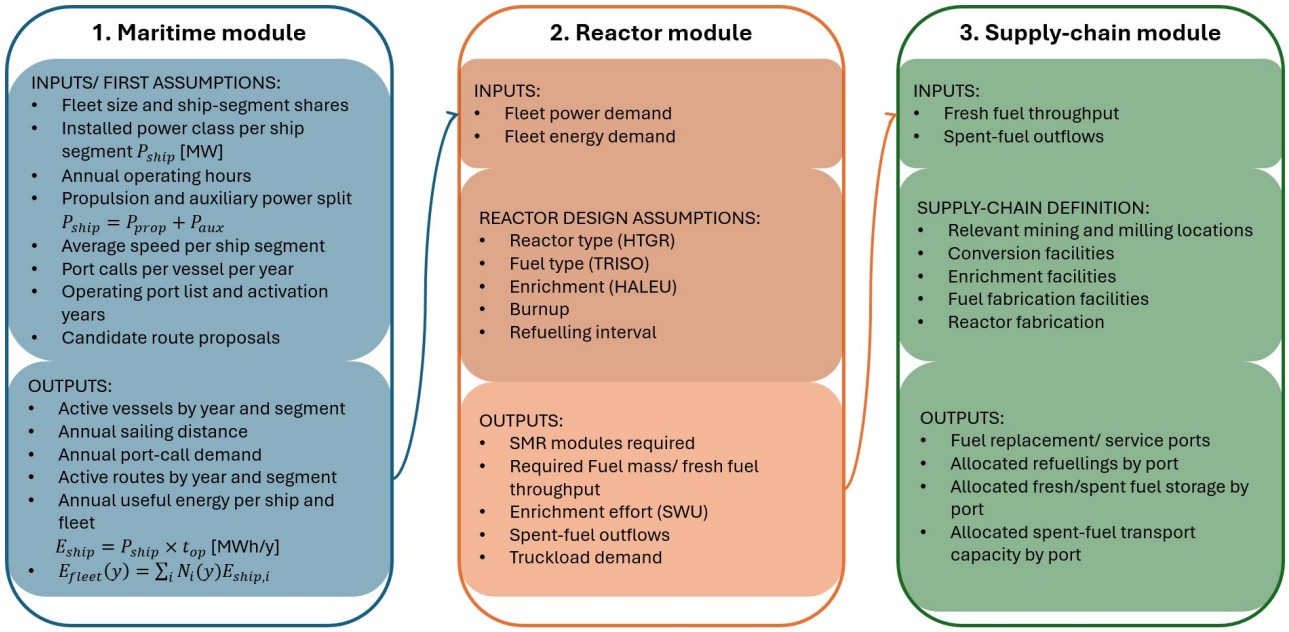


Figure 1: Integrated model structure showing the connection between the maritime fleet deployment, route-network allocation, reactor and fuel-demand calculations, global fuel-cycle infrastructure requirements, and localized nuclear port-service hub allocation.

2.1 Maritime module

The maritime module defines the representative nuclear-powered fleet considered in the baseline scenario and translates the fleet assumptions into the quantities required by the rest of the model. Its main outputs are the number of active vessels and maritime routes by ship segment and year, the corresponding installed power demand and annual energy demand. These outputs are used to design the route network and provide the main inputs for the reactor and supply-chain modules.

2.1.1 Ship segments and fleet composition

The baseline scenario considers a fleet of $N_{\text{ships}} = 800$ nuclear-powered vessels by 2060. Based on the 2025 Allseas SMR impact study [2], the fleet is grouped according to representative installed electrical-power requirements. Four power classes are used in the model: 25, 50, 75 and 100 MWe, corresponding to one, two, three or four 25 MWe SMR units, respectively. These power classes are then associated with the ship segments shown in Table 1.

The selection of ship segments is informed by the maritime SMR deployment considerations reported in the 2025 Allseas SMR impact study [2] (see Table 2). The Allseas study does not provide a fleet-composition scenario for this project, but it identifies which vessel categories are more or less attractive for SMR adoption. In particular, offshore vessels, container ships and bulk carriers are highlighted as relevant candidates because of their operational profiles, fuel demand, route characteristics or potential benefit from long refuelling cycles. These values are therefore interpreted here as qualitative deployment indicators rather than as direct percentages to be applied to the 800-ship fleet.

Table 3 shows the resulting baseline fleet composition for the model. The container-ship share is divided between Panamax, large and ultra-large containerships, with a larger fraction assigned to the higher-power classes because container ships are identified by Allseas as particularly attractive candidates for SMR adoption

Table 1: Representative ship segments for different installed propulsion power levels using 25 MWe SMR units. Adapted from Allseas SMR impact study [2].

Installed power (MWe)	**Number of 25 MWe SMRs**	**Representative segments**
25	1	Mid-size bulk carriers; offshore supply vessels; medium passenger ferries; large fishing vessels; chemical tanker
50	2	Panamax containerships; mid-size icebreakers
75	3	Large containerships; large oil tankers
100	4	Ultra-large containerships; large icebreakers; Large cruise ships

Table 2: Maritime SMR deployment considerations by vessel type in 2025. Adapted from Allseas SMR impact study [2].

Vessel type	**Considerations for SMR deployment**	**Relative SMR adoption score**
Bulk carrier	Low speeds and fuel use reduce potential fuel-cost savings from SMR adoption.	37.5%
Container ship	High speeds and fixed long-distance routes create strong potential for fuel-cost savings.	50%
Offshore vessel	Remote operations could benefit from long SMR fuel cycles.	62.5%
Oil tanker	Lower decarbonization pressure may limit adoption, despite possible fuel-cost savings.	25%
Other	This category includes a wide range of ship segments, with different speeds and fuel use.	37.5%

due to their high speeds and fixed long-distance routes. The "Other" category is used as a proxy for vessel types that are included in the model but not explicitly separated in the Allseas deployment table, such as icebreakers and fishing vessels.

2.1.2 Annual nuclear-service operating time

The annual operating time of each ship segment is defined as the number of hours per year during which the vessel is assumed to be in nuclear-service operation and therefore requires propulsion power. Time spent in port and at anchorage is not included in this operating time, although auxiliary-power demand may still be relevant during those periods.

The 2024 European Commission report on CO_2 emissions from maritime transport provides average time-at-sea values by ship type within the scope of the EU Maritime MRV Regulation [3]. These values are used as lower reference values, but they are not directly adopted as the operating hours of the nuclear fleet. The reason is that the MRV data only represent monitored activity for ships sailing to, from or between ports in the European Economic Area, whereas the baseline nuclear fleet considered in this study is assumed to operate globally. In addition, nuclear-powered vessels are assumed to be preferentially deployed on long-distance or high-utilisation routes, where the high capital cost of the reactor system and the long refuelling interval make intensive operation more favourable. For this reason, the model introduces a higher set of nuclear-service operating hours, $t_{\mathrm{op,nuc}}$, which should be interpreted as scenario assumptions for a mature global nuclear fleet rather than observed historical averages.

Table 4 compares the European Commission reference values with the operating hours adopted in the nuclear-fleet model. For example, the adopted value of 5,000 h/year for bulk carriers corresponds to about 57% of the year, or approximately 208 full operating days. Although this is more than three times the European Commission MRV reference value, the difference is justified because of the reasons explained before. Similarly, the 6,500 h/year assumed for container ships corresponds to about 74% of the year, or approximately 271 full operating days, reflecting their regular long-distance route structure and the strong economic incentive to maximise the utilisation of high-capital-cost nuclear vessels. Passenger ferries are assigned the highest value, 6,800 h/year, because they are assumed to operate on frequent scheduled services, while icebreakers and fishing vessels are assigned lower values of 4,500 h/year because their operation is more seasonal or geographically constrained. All adopted values remain below 8,760 h/year, leaving explicit time for port stays, maintenance, inspections and operational downtime.

Table 3: Baseline fleet-composition assumptions by ship segment for an 800-ship fleet for year 2060.

Model ship segment	**Allseas category**	**Installed power (MWe)**	**Fleet share (%)**	**Ships in 800-ship fleet**
Bulk carrier	Bulk carrier	25	12.5	100
Panamax container	Container ship	50	7.0	56
Large container	Container ship	75	18.0	144
Ultra-large container	Container ship	100	20.0	160
Offshore vessel	Offshore vessel	25	21.6	173
Oil tanker	Oil tanker	75	2.8	22
Chemical tanker	Other	25	2.5	20
Medium icebreaker	Other	50	3.0	24
Large icebreaker	Other	100	3.0	24
Fishing vessels	Other	25	2.1	17
Large cruise	Other	100	2.5	20
Passenger ferry	Other	25	5	40
Total Ships	–	–	**100.0**	**800**
Total SMR	–	–		**1824**

Table 4: Reference and adopted annual operating hours by ship segment. The European Commission values are approximate 2023 average time-at-sea values within the EU Maritime MRV scope [3], while the nuclear-fleet values are scenario assumptions for global high-utilisation operation.

Model ship segment	**EC MRV ship type**	**EC MRV reference t_{op} [h/year]**	**Adopted nuclear-fleet $t_{\mathrm{op,nuc}}$ [h/year]**
Bulk carrier	Bulk carrier	$\sim$ 1,400	5,000
Panamax container	Container ship	$\sim$ 2,700	6,500
Large container	Container ship	$\sim$ 2,700	6,500
Ultra-large container	Container ship	$\sim$ 2,700	6,500
Offshore vessel	Other ship types	$\sim$ 2,350	5,500
Oil tanker	Oil tanker	$\sim$ 2,100	5,500
Chemical tanker	Chemical tanker	$\sim$ 2,200	5,500
Medium icebreaker	Other ship types	$\sim$ 2,350	4,500
Large icebreaker	Other ship types	$\sim$ 2,350	4,500
Fishing vessels	Other ship types	$\sim$ 2,350	4,500
Large cruise	Passenger ship	$\sim$ 2,900	6,200
Passenger ferry	Ro-Pax ship	$\sim$ 4,100	6,800

2.1.3 Propulsion power, auxiliary power and vessel speed

The installed electrical-power classes of 25, 50, 75 and 100 MWe are retained from the Allseas maritime SMR concept [2]. However, installed power is interpreted as nameplate electrical capacity and is not assumed to represent the average power demand of the vessel. In normal operation, only a fraction of the installed capacity is used for propulsion and auxiliary services.

For each ship segment i, a propulsion fraction is defined from reference conventional vessels as

$$f_i = \frac{P^{\mathrm{ref}}_{\mathrm{inst},i}}{T^{\mathrm{ref}}_{\mathrm{inst},i}}, \tag{1}$$

where $P^{\mathrm{ref}}_{\mathrm{inst}}$ is the installed propulsion power and $T^{\mathrm{ref}}_{\mathrm{inst}}$ is the corresponding total installed power. Thus, $A^{\mathrm{ref}}_{\mathrm{inst}} = T^{\mathrm{ref}}_{\mathrm{inst}} - P^{\mathrm{ref}}_{\mathrm{inst}}$ in the installed auxiliary power. Reference values for $T^{\mathrm{ref}}_{\mathrm{inst}}$, $P^{\mathrm{ref}}_{\mathrm{inst}}$ and $A^{\mathrm{ref}}_{\mathrm{inst}}$ are taken from CE Delft representative vessel designs [4].

An average propulsion load factor L_i is defined as follows for each ship segment i:

$$L_i = \frac{P^{\mathrm{ref}}_{\mathrm{use},i}}{P^{\mathrm{ref}}_{\mathrm{inst},i}} = \frac{P^{\mathrm{mod}}_{\mathrm{use},i}}{P^{\mathrm{mod}}_{\mathrm{inst},i}} \tag{2}$$

Here $P^{\mathrm{ref}}_{\mathrm{use},i}$ is the average propulsion power usage for reference ship segment i during operation. $P^{\mathrm{mod}}_{\mathrm{use},i}$, on the other hand, is the average propulsion power usage for ship segment i in the model. Finally, $P^{\mathrm{mod}}_{\mathrm{inst},i}$ is the propulsion installed power for ship segment i in the model following the Allseas power classes. Thus, L_i represents the fraction of P_{inst} that is actually being used for propulsion in operation on average. The load factor L_i is calculated from the reference values and used to estimate $P^{\mathrm{mod}}_{\mathrm{use},i}$ of the model's ship segments following Eq. (3).

$$P^{\mathrm{mod}}_{\mathrm{use},i} = T^{\mathrm{mod}}_{\mathrm{inst},i}\, f_i\, L_i. \tag{3}$$

The installed auxiliary power of the modelled ship segments is obtained from the remaining fraction of the total installed electrical capacity,

$$A^{\mathrm{mod}}_{\mathrm{inst},i} = T^{\mathrm{mod}}_{\mathrm{inst},i} - P^{\mathrm{mod}}_{\mathrm{inst},i}. \tag{4}$$

The auxiliary load factor is obtained independently from the reference vessel,

$$L_{A,i} = \frac{A^{\mathrm{ref}}_{\mathrm{use},i}}{A^{\mathrm{ref}}_{\mathrm{inst},i}}, \qquad A^{\mathrm{mod}}_{\mathrm{use},i} = L_{A,i} A^{\mathrm{mod}}_{\mathrm{inst},i}. \tag{5}$$

When available, load factors L_i and $L_{A,i}$ are calculated directly from CE Delft power split values [4]. For the remaining segments, the load factors are estimated.

For simplicity, the cubic speed–power approximation is applied to all ship segments in the present model:

$$P^{\mathrm{mod}}_{\mathrm{use}}(v) = P^{\mathrm{ref}}_{\mathrm{use}} \left(\frac{v}{v_{\mathrm{ref}}} \right)^3, \tag{6}$$

which gives the modelled average speed

$$v_{\mathrm{avg},i} = v_{\mathrm{ref},i} \left(\frac{P^{\mathrm{mod}}_{\mathrm{use},i}}{P^{\mathrm{ref}}_{\mathrm{use},i}} \right)^{1/3}. \tag{7}$$

Average used auxiliary power A_{use} is specified independently using representative at-sea auxiliary-engine loads from CE Delft [4]. Consequently, $P_{\mathrm{use},i} + A_{\mathrm{use},i}$ is generally lower than the installed total electrical capacity $T_{\mathrm{inst},i}$ for all the ship segments.

Table 5 shows the reference values for each ship segment considered in this model. These reference values will be used to calculate the ultimate values used in the model.

Table 5: Reference power and speed parameters used to derive the maritime-model inputs. f_{prop} is the rated propulsion-to-installed power ratio and L_{prop} is the adopted average propulsion load factor. Auxiliary power represents average at-sea demand.

Ship segment	f	L	$T^{\mathrm{ref}}_{\mathrm{inst}}$ [MW]	$P^{\mathrm{ref}}_{\mathrm{inst}}$ [MW]	$P^{\mathrm{ref}}_{\mathrm{use}}$ [MW]	$A^{\mathrm{ref}}_{\mathrm{inst}}$ [MW]	$A^{\mathrm{ref}}_{\mathrm{use}}$ [MW]	v_{ref} [kn]
Bulk carrier	0.85	0.56	9.7	8.20	4.56	1.50	0.83	14.40
Panamax container	0.71	0.41	34.60	24.70	10.06	9.90	4.03	23.10
Large container	0.71	0.57	60.20	43.00	24.57	17.20	9.83	23.90
Ultra-large container	0.71	0.57	90.00	63.90	36.42	26.1	14.88	20.00
Offshore vessel	0.85	0.56	9.7	8.20	4.56	1.5	0.83	11.00
Oil tanker	0.79	0.47	27.10	21.40	10.14	5.7	2.70	14.60
Chemical tanker	0.79	0.50	27.10	21.40	10.70	5.7	2.85	13.30
Medium icebreaker	0.71	0.41	34.60	24.70	10.06	9.90	4.03	11.90
Large icebreaker	0.71	0.57	60.20	43.00	24.57	17.20	9.83	10.10
Fishing vessel	0.71	0.41	34.60	24.70	10.06	9.90	4.03	15.00
Large cruise	0.70	0.73	76.1	53.27	38.89	22.83	16.67	22.50
Passenger ferry	0.70	0.73	38.05	26.64	19.44	11.41	8.33	24.00

2.1.4 Final maritime input parameters

The resulting ship-segment input assumptions are summarized in Table 6. Each row corresponds to one ship-segment instance used by the fleet model. The table includes the installed electrical power, the estimated split between propulsion and auxiliary power, the representative average speed, the annual operating time, and the assumed number of port calls per year.

The port-call assumptions are chosen according to the expected service pattern of each vessel segment. Bulk carriers, tankers and icebreakers are assigned relatively low values, between 15 and 20 port calls per year, because they are assumed to perform longer voyages with fewer port visits. Container ships are assigned intermediate

Table 6: Final ship-segment input assumptions used in the maritime module. Installed power retains the Allseas SMR power classes, while propulsion power, auxiliary power and speed are derived using the reference parameters in Table 5 and equations (3) and (7).

Ship segment	$T_{\text{inst}}^{\text{mod}}$ [MW]	$P_{\text{inst}}^{\text{mod}}$ [MW]	$P_{\text{use}}^{\text{mod}}$ [MW]	$A_{\text{inst}}^{\text{mod}}$ [MW]	$A_{\text{use}}^{\text{mod}}$ [MW]	v_{avg} [kn]	t_{op} [h/year]	Port calls/year
Bulk carrier	25	21.13	11.75	3.87	2.14	19.74	5,000	15
Panamax container	50	35.69	14.54	14.31	5.82	26.12	6,500	40
Large container	75	53.57	30.61	21.43	12.25	25.72	6,500	50
Ultra-large container	100	71.00	40.47	29.00	16.53	20.71	6,500	45
Offshore vessel	25	21.13	11.75	3.87	2.14	15.08	5,500	50
Oil tanker	75	59.01	27.96	15.99	7.58	20.47	5,500	20
Chemical tanker	25	19.67	9.83	5.33	2.67	12.93	5,500	40
Medium icebreaker	50	35.69	14.54	14.31	5.82	13.45	4,500	20
Large icebreaker	100	71.43	40.81	28.57	16.33	11.96	4,500	15
Fishing vessel	25	17.85	7.27	7.15	2.91	13.46	4,500	40
Large cruise	100	70.00	51.10	30.00	21.91	24.64	6,200	100
Passenger ferry	25	17.50	12.77	7.50	5.47	20.86	6,800	500

values, between 40 and 50 port calls per year, reflecting regular liner operation with repeated calls at major route nodes. Offshore vessels and fishing vessels are also assigned intermediate values because their operation involves more frequent returns to shore or service bases. Passenger vessels are assigned higher values: large cruise ships are assumed to make 100 port calls per year, while passenger ferries are assigned 500 port calls per year because they represent frequent scheduled services. These port calls should be interpreted as ordinary maritime port calls, not as nuclear refuelling events.

2.2 Reactor module

The reactor module connects the maritime fleet assumptions defined in Section 2.1 with the fuel-cycle and infrastructure calculations developed later in the report. This study uses a pebble-bed high-temperature gas-cooled reactor (HTGR) concept as the reference shipboard reactor. The reference design is based on the maritime SMR concept discussed in the Allseas SMR impact study [2]. In that study, the reactor is described as a 25 MWe module with a thermal power of approximately 70 MWt, using helium coolant and TRISO fuel with uranium oxycarbide (UCO) kernels. These characteristics are used here to define the representative reactor unit of the model. The reactor module should therefore be interpreted as a generic maritime pebble-bed HTGR module with similar power, fuel type and refuelling behaviour, rather than as a detailed engineering model of the Allseas design.

The Allseas SMR impact study does not specify the fuel enrichment, burnup or detailed neutronic design of the proposed maritime HTGR. These parameters must therefore be assumed from comparable pebble-bed or TRISO-fuelled HTGR concepts. The reference designs in Table 7 show that enrichment levels vary significantly between existing and proposed HTGRs. The HTR-PM uses fuel with comparatively lower enrichment than the compact advanced concepts Xe-100 and BANR. However, all three concepts use fuel enrichment in the HALEU range. Given the compact size of the maritime reactor and its power level, it is reasonable to assume that the Allseas-type reactor would also require HALEU fuel.

Table 7: Reference HTGR concepts used to guide the reactor assumptions of the maritime model. Data from HTR-10 [5], HTR-PM [6], Xe-100 [7] and BANR [8].

Concept	Thermal power [MWt]	Fuel type	Enrichment	Burnup [GWd/tU]
HTR-10	10	UO_2	17%	80
HTR-PM	250	–	8.6%	–
Xe-100	200	UCO	15.5%	168
BANR	75	–	19.75%	–

In this model, the fuel enrichment is set to

$$x_p = 15\%.$$

This assumption is comparable with other TRISO-fuelled reactor concepts such as Xe-100, while lower than the BANR [7, 8]. The use of HALEU is important for the infrastructure analysis because uranium enriched above 5% may require stricter security, licensing and handling requirements than conventional low-enriched uranium, at least in the US [9].

The model also assumes that the reactor operates for five years between refuellings. This refuelling interval requires a sufficiently large initial fuel loading and enough excess reactivity to sustain operation over the full

cycle. Since the detailed neutronics of the Allseas reactor are unknown, this study treats the five-year cycle as a scenario assumption rather than a verified reactor-design result.

To calculate the corresponding fuel mass, a representative burnup value is required. In the baseline model, the burnup is set to

$$BU = 80 \text{ GWd/tU}.$$

This value is consistent with the HTR-10 reference value shown in Table 7, while being less than half of the reported value for the Xe-100. The lower of the two values was chosen as a conservative estimate.

3 Supply chain

3.1 Reactor manufacturing context

Building a nuclear-powered shipping fleet would require an industrial supply chain capable of manufacturing, assembling, testing, delivering and maintaining the required shipboard reactor modules. In the baseline fleet composition shown in Table 3, the final 2060 fleet requires 1884 reactor modules of 25 MWe each. Reactor manufacturing could therefore become a major supply-chain bottleneck in a real deployment scenario.

The reference reactor concept used in this study is the Allseas maritime HTGR SMR described in Section 2.2. Allseas states that its nuclear roadmap includes design studies and pre-licensing in 2025–2026, basic and detailed design and testing in 2027–2028, the start of production at a dedicated facility in 2029–2030, and initial deployment on land followed by offshore roll-out from 2030 [10]. This supports the assumption that the first reactor modules could become available before the maritime deployment period considered in this report.

However, the present model does not include a quantitative reactor-manufacturing module. It does not assign annual reactor-production capacities, factory construction times, certification delays, component supply limits or transport constraints. Instead, the model assumes that the reactor modules required by the fleet deployment are available when needed, and then focuses on the downstream infrastructure consequences: HALEU enrichment, TRISO fuel fabrication, refuelling demand, spent-fuel generation, storage, transport and nuclear-capable port-service capacity.

This simplification is important for interpreting the results. Reactor manufacturing is acknowledged as a possible bottleneck, but it is not one of the bottlenecks quantified in the present version of the model. The quantitative results presented later in the report therefore refer to fuel-cycle and port-infrastructure requirements, not to the feasibility of manufacturing 1824 reactor modules by 2060.

A more complete future version of the model could introduce a dedicated reactor-supply module. Such a module could distribute reactor manufacturing, assembly, testing and maintenance capacity across several regions. Although Allseas currently lists fabrication facilities in the Netherlands, it also has engineering offices, logistics facilities and project offices in other regions, including Southeast Asia, the Middle East, North America, South America and Oceania [11]. These locations could provide a qualitative basis for defining future regional reactor-supply nodes. This extension would allow the model to test whether the assumed fleet-deployment curve is compatible not only with fuel-cycle and port-service capacity, but also with reactor-production capacity.

A qualitative example of how such a future reactor-supply module could be regionalised is provided in Appendix B.

3.2 Fuel

The reactor module defined in Section 2.2 assumes a maritime pebble-bed HTGR using HALEU fuel and TRISO particles with UCO kernels. The fuel supply chain considered in this study must therefore provide enriched uranium in the HALEU range and the specific fabrication steps required to produce TRISO fuel elements. This section defines the simplified fuel-cycle scope used in the model and identifies the enrichment and fuel-fabrication capacities assumed for the baseline scenario. The general development of advanced reactor fuel cycles, including the need to establish suitable front-end and back-end infrastructure, is discussed in the National Academies fuel-cycle assessment [12].

HALEU is defined as uranium enriched above 5% and below 20% in ^{235}U [13]. HALEU can be produced either by downblending higher-enriched uranium or by enriching lower-grade uranium. Since civilian industry does not generally have access to highly enriched uranium, this study assumes that HALEU production occurs through enrichment of lower-enriched uranium using different methods such as gas centrifugation. In addition to enrichment capacity, the deployment of a maritime HTGR fleet requires TRISO fuel-fabrication capacity. This is treated separately from conventional light-water-reactor fuel fabrication because TRISO production requires fuel-kernel fabrication, coating of the particles and incorporation of the coated particles into fuel elements such as pebbles or compacts [14].

The fuel-cycle model is simplified to a once-through HALEU–TRISO fuel cycle, as shown in Figure 2. The first stages, uranium mining and milling and uranium conversion, are included to represent the upstream supply chain. The stages that are treated most explicitly in the capacity assumptions are HALEU enrichment and TRISO fuel fabrication, because these are the steps most directly affected by the reactor assumptions and most likely to become bottlenecks for a maritime HTGR fleet. The fuel-cycle scheme is adapted from the general advanced fuel-cycle structure presented in the Advanced Fuel Cycle Cost Basis report [15].

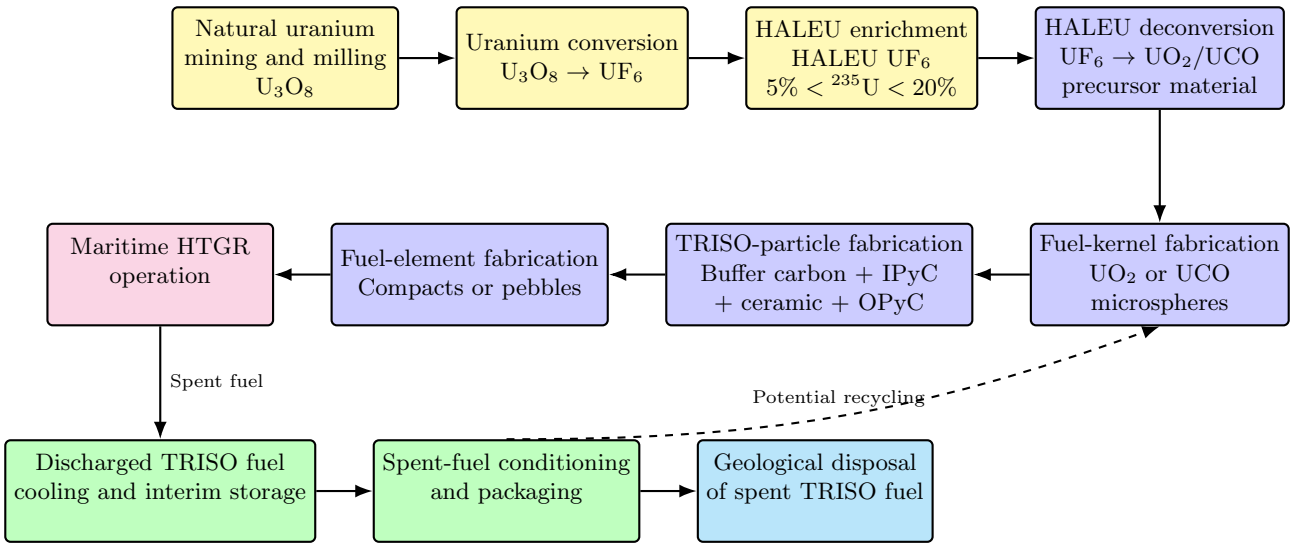


Figure 2: Proposed once-through HALEU–TRISO fuel-cycle scheme for maritime high-temperature gas-cooled reactors, adapted from the general fuel-cycle module structure in Figure S-1 of the *Advanced Fuel Cycle Cost Basis* [15].

Figures 3 and 4 show the geographical distribution of uranium mine production and uranium conversion capacity, respectively [16]. These figures are used to provide context for the upstream part of the fuel cycle.

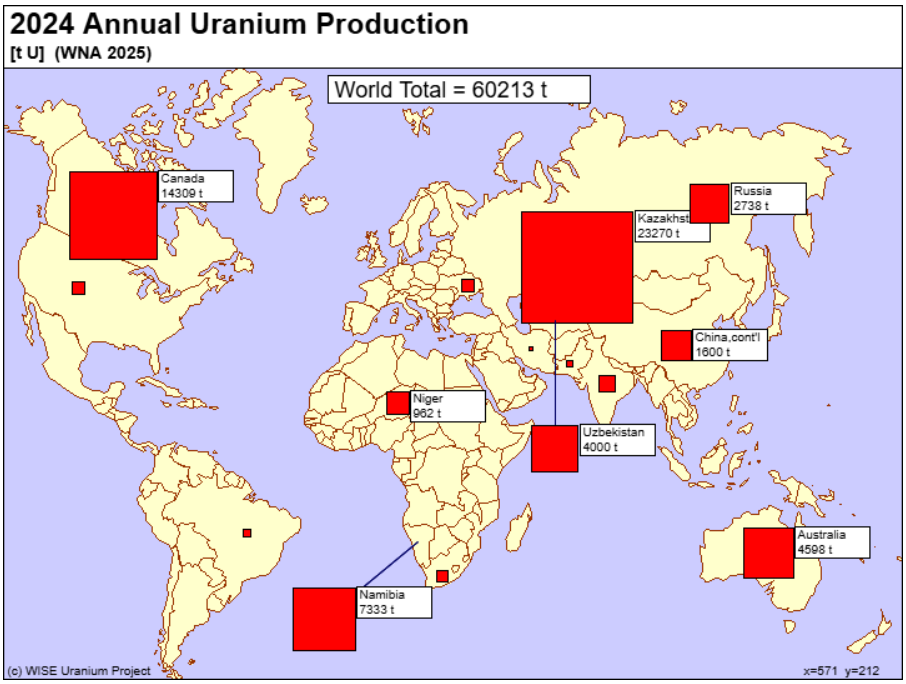


Figure 3: Annual uranium mine production by country in 2024, expressed as tonnes of contained uranium (tU). These values represent the uranium content of the concentrate produced at the mining and milling stage [16].

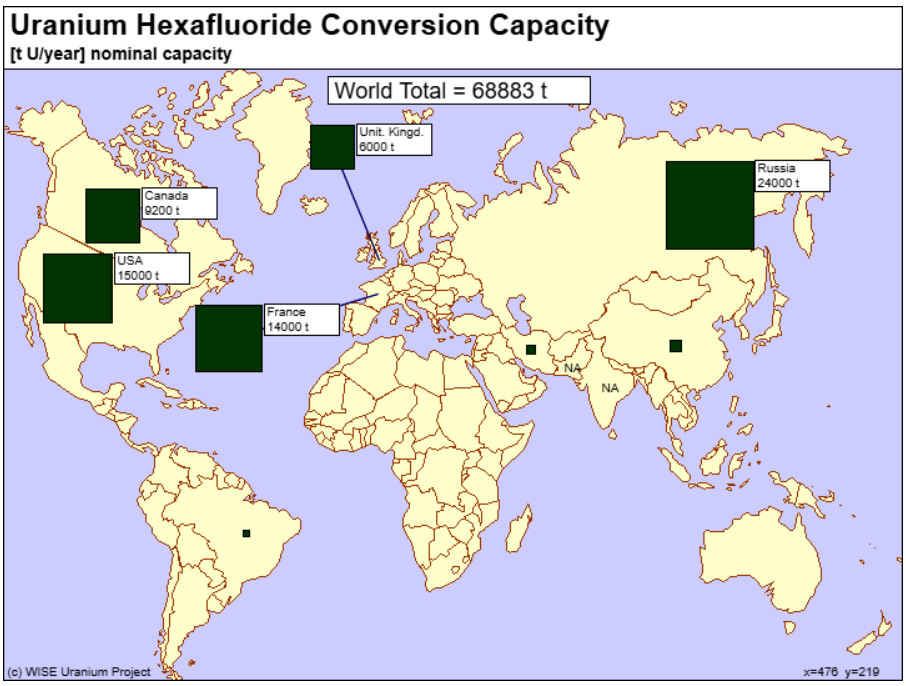


Figure 4: Uranium Hexafluoride Conversion Capacity in 2024, expressed as tonnes of contained uranium (tU). [16].

For HALEU enrichment, the following regional assumptions are used. The Urenco company in the UK is expected to have HALEU production capability by the early 2030s. According to the Euratom Supply Agency Annual Report, Urenco will have the capacity to produce 10 tU/year of HALEU by 2031 [17]. However, in this model Urenco HALEU enrichment capacity is represented as 27 tU/year from 2035, based on Urenco's advanced-fuels information [18]. Russian HALEU enrichment through Tenex is treated as commercially available at industrial scale in qualitative terms, although its annual HALEU output capacity is not publicly specified [19]. The model therefore assigns Russia a proxy value of 30 tU/year from 2035. In the United States, Centrus is treated as the only facility actually producing HALEU at present. Centrus produced 900 kg of HALEU by June 2025, so the current U.S. HALEU enrichment capacity is approximated as 1 tU/year in the model from 2035 [20]. China is included as a domestic HALEU-capable country because public sources identify China, together with Russia, as one of the countries with infrastructure to produce HALEU at scale [19]. Since no specific Chinese HALEU capacity in tU/year is publicly reported, the model assigns China a proxy HALEU enrichment capacity of 20 tU/year from 2035.

For TRISO fuel fabrication, the United States is represented by the TRISO-X TX-1 facility. The facility is reported to have an initial production capacity of 5 tU/year and a potential expanded capacity of 16 tU/year by the early 2030s [21]. The model therefore assigns the United States a TRISO fabrication capacity of 16 tU/year from 2035. However, because this fuel uses HALEU, this fabrication capacity can only be fully used if sufficient HALEU feedstock is available. Under the baseline U.S. HALEU enrichment assumption of 1 tU/year, the United States would therefore require imported HALEU or additional domestic enrichment capacity to operate the full TRISO-X fabrication capacity.

Japan is included as a potential TRISO fuel-fabrication country because Nuclear Fuel Industries successfully fabricated the first loading fuel for the HTTR, including UO_2 kernels, TRISO-coated fuel particles, fuel compacts and fuel rods [22]. The HTTR campaign completed about 900 kgU of first-loading fuel, while the NFI HTR fuel plant had a licensed capacity of 400 kgU/year. More importantly for scale-up assumptions, the same reference states that two lines of kernel and coated-particle fabrication equipment could produce about 5 tU/year, and that modular installation could support a 10 tU/year HTR fuel plant [22]. The model does not assume that Japan currently produces the exact HALEU–TRISO fuel required for the maritime HTGR scenario. Instead, Japan is treated as a candidate fabrication country with demonstrated HTR/TRISO manufacturing capability. A reactivated TRISO fuel-fabrication proxy of 10 tU/year is assigned from 2040.

China is included as both HALEU-capable and TRISO-capable. For TRISO fuel fabrication, the HTR-PM fuel production line is used as the reference. Its reported production capacity is 300,000 spherical fuel elements per year [23]. Since each HTR-PM pebble contains approximately 7 g of enriched fuel, this corresponds to about 2.1 tU/year. This value is used as the baseline Chinese TRISO fabrication proxy from 2035, but it is interpreted as adaptable capacity rather than as a confirmed production line for the exact Allseas fuel specification. Producing the project-specific HALEU–TRISO fuel would require fuel-design adaptation, qualification and licensing.

France is included as a potential future European TRISO fuel-fabrication country through Framatome's Romans-sur-Isère site. Framatome announced new industrialisation and development capabilities for TRISO fuel in France, and is planning to produce 2 tU/year of HALEU TRISO fuel from 2027 in partnership with Standard Nuclear [24]. In the baseline model, France is therefore treated as a European TRISO fabricator with a capability of 2 tU/year from 2035.

The resulting baseline enrichment and TRISO fabrication assumptions are summarised in Table 8.

Table 8: Projected HALEU and TRISO production capacity in 2035.
a: Additional 10 tU/year in Japan from 2040.

	OECD (excl. USA)	USA	Russia	China	Total
HALEU [tU/year]	27	1	30	20	78
TRISO [tU/year]	2 (12)[a]	16		2.1	20.1 (30.1)[a]

4 Country profiles

4.1 Country-level roles in the model

The country profiles provide the link between the supply-chain context described in Section 3 and the maritime route network introduced in the following subsection. Each country in the model can have one or more roles. First, a country may contribute to the reactor or fuel supply-chain context. This includes reactor manufacturing, HALEU enrichment and TRISO fuel fabrication. Second, a country may contribute to the maritime operating network through one or more ports. These two roles are defined independently: a country can be a supply-chain country without operating ports, an operating-port country without supply-chain capacity, or both.

Table 9 provides a compact description of how each country is represented in this scenario model. The reactor-manufacturing entries in Table 9 should be interpreted as qualitative scenario assumptions based on the possible future regionalisation discussed in Appendix B. They are not used as numerical inputs in the present model. Instead, it assumes that the reactor modules required by the fleet deployment are available when needed, and focuses the quantitative analysis on maritime deployment, fuel-cycle requirements, storage, transport and port-service capacity.

The country input file therefore acts as a common database for the integrated model. For each country, it specifies whether the country is included as a supply-chain participant, which supply-chain step is represented, the assumed start year of that role, whether the country contributes operating ports, and the year from which each port enters the maritime network. In the numerical maritime model, the operating-port information is the most important part of this database: it is used to construct the maritime route network.

Table 9: Country-level roles used in the integrated maritime nuclear infrastructure model.

Country	**Supply-chain role in the model**	**Start year**	**Maritime operating role**
Netherlands	Reactor manufacturing / early European reactor-supply base	2030	Operating ports in North-West Europe.
Malaysia	Regional reactor manufacturing for Southeast Asia	2038	Operating ports in the Malacca / Southeast Asian corridor.
United Arab Emirates	Regional reactor manufacturing for the Middle East	2041	Jebel Ali is included as a major route hub.
United States	HALEU enrichment, TRISO fuel fabrication, and later Regional reactor manufacturing	2035 / 2047	Operating ports on the Pacific and Atlantic coasts.
Brazil	Regional reactor manufacturing for South America	2054	Santos extends the operating network to the South Atlantic.
Australia	Regional reactor manufacturing for Oceania	2055	Melbourne and Newcastle extend the operating network to Oceania.
United Kingdom	HALEU enrichment	2035	No operating port is included in the baseline route network.
Russia	HALEU enrichment	2035	Arctic operating ports are included for the icebreaker subnetwork.

Country	Supply-chain role in the model	Start year	Maritime operating role
China	HALEU enrichment and TRISO fuel fabrication	2035	Several major Chinese ports are included in the Asian operating network.
France	TRISO fuel fabrication	2035	French ports are potential ports in the European and Mediterranean operating network.
Japan	TRISO fuel-fabrication	2040	No operating port is included in the baseline route network.
Other operating-port countries	No supply-chain role in the baseline model	–	Included through maritime ports only. These include, among others, Belgium, Germany, Singapore, South Korea, Spain, Norway, Poland, Romania, Italy, Taiwan, Thailand and South Africa.

4.2 Port and route geography

The candidate port set is selected to represent both commercial relevance and strategic geographical coverage (see Table 10). Some of the global ports included in the model, specially those in Asia, are among the major container ports worldwide and therefore represent high-volume nodes for long-distance maritime routes [25]. The European ports are taken from the top European ports by annual cargo tonnage. Additional ports in Brazil, Australia, South Africa and North America are included because of their strategic geographical position. Therefore, the port list is not intended to represent all possible nuclear-ready ports, but rather a representative set of high-relevance route nodes for testing the geographical expansion of the fleet operation scenario.

Although the baseline port network is mainly based on major container ports, major European cargo ports and additional strategic ports, this network does not fully represent the operating geography of icebreakers. Shu et al. [26] show that icebreaker-assisted navigation is particularly relevant in ice-covered waters of the Northern Sea Route (NSR), which connects the Atlantic and Pacific Oceans along the Russian Arctic coast and forms part of the Northeast Passage. Therefore, the model introduces an additional Arctic operating subnetwork for icebreaker segments: Port of Murmansk, Port of Arkhangelsk and Port of Provideniya in Russia.

Table 10: Assumed port-network for the baseline fleet-operation scenario, including activation years. The port selection includes the major container ports worldwide [25].

Port	Country / region	Start year
Port of Rotterdam	Netherlands	2035
Port of Antwerp-Bruges	Belgium	2035
Port of Hamburg	Germany	2035
Port of Singapore	Singapore	2035
Port of Shanghai	China	2035
Port of Busan	South Korea	2038
Port of Jebel Ali	UAE	2039
Port of Los Angeles	USA	2040
Port of Provideniya	Russia	2041
Port of Arkhangelsk	Russia	2041
Port of Long Beach	USA	2041
Port of Ningbo-Zhoushan	China	2043
Port of Savannah	USA	2043
Port of Shenzhen	China	2044
Port of Hong Kong	China	2045
Port Klang	Malaysia	2045
Port of Tanjung Pelepas	Malaysia	2045
Port of Qingdao	China	2045
Port of Guangzhou	China	2046
Port of Tianjin	China	2046
Port of Xiamen	China	2046

Port	Country / region	Start year
Port of Murmansk	Russia	2046
Port of New York and New Jersey	USA	2046
Port of Kirkenes	Norway	2047
Port of Kaohsiung	Taiwan	2047
Port of Laem Chabang	Thailand	2047
Port of Algeciras	Spain	2048
HAROPA / Le Havre-Rouen-Paris	France	2048
Port of Valencia	Spain	2048
Port of Tromsø	Norway	2048
Port of Marseille-Fos	France	2049
Port of Barcelona	Spain	2049
North Sea Port	Netherlands	2049
Ports of Bremen / Bremerhaven	Germany	2050
Seaports of Niedersachsen	Germany	2050
Port of Gdańsk	Poland	2050
Port of Genoa	Italy	2051
Port of Trieste	Italy	2051
Port of Constanța	Romania	2052
Port of Bergen	Norway	2052
Port of Santos	Brazil	2054
Port of Melbourne	Australia	2055
Port of Newcastle	Australia	2056
Port of Durban	South Africa	2057

5 Deployment

5.1 Fleet deployment scenario

The deployment of the fleet is assumed to follow an S-curve, described by the sigmoid function

$$\sigma(x) = \frac{1}{1 + \exp(-a(x - x_0))},$$

where x is the year after 2035, a is the steepness of the curve and x_0 is the midpoint of the deployment interval, i.e. $x_0 = (2060 - 2035)/2$. The steepness of the deployment curve will affect the peak demands for fuel storage and port capacities.

The deployment can be divided into two distinct phases. In phase 1, deployment is slow. The supply chains are under development and there is no common international regulatory framework. The small nuclear fleet is confined to shipping routes between countries with bilateral or multilateral agreements.

After the feasibility of the nuclear shipping fleet has been proven and an international regulatory framework is in place, phase 2 begins. Here, there is a period of rapid ship deployment before market saturation is reached. This also presupposes that reactor and fuel production, along with expansions in port infrastructure, can meet the demand. One can assume that phase 1 is dominated by different types of cargo ships, while the passenger-carrying ships will only start deployment in phase 2.

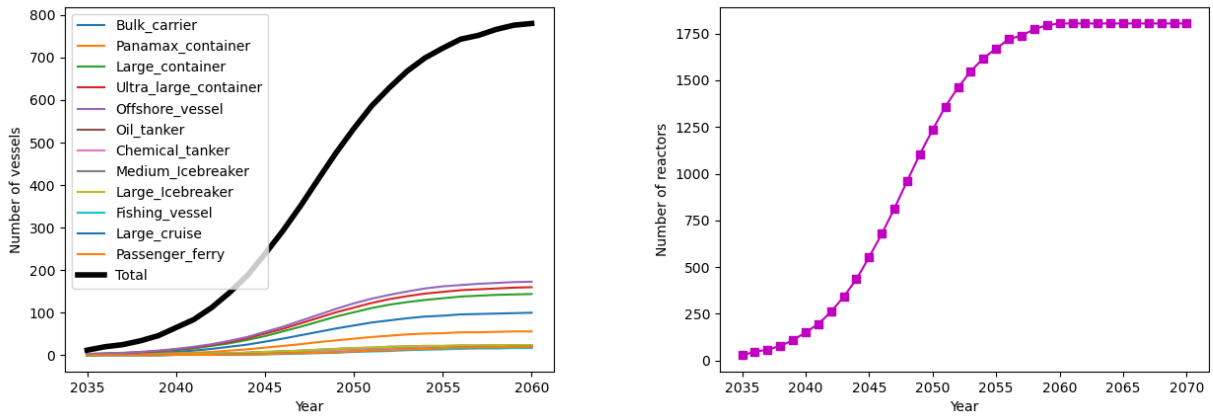
For each year after 2035, the number of vessels in a ship segment is calculated using a sigmoid function multiplied by the number of ships present in the fleet in 2060, as described by Table 3. The postponement of the passenger-carrying ship deployment to phase 2 is handled by defining two separate sigmoid functions. For the ships present in phase 1, the following function is used to calculate the number of ships in year x after 2035,

$$n_1(x) = \frac{K_1}{1 + \exp(-a(x - x_0))},$$

where K_1 is the number of vessels in a ship segment in 2060. For the ships postponed to phase two, the midpoint is shifted by $x_1/2$ years, so that its midpoint is given by $\tilde{x}_0 = (2060 - 2035 + x_1)/2$. The number of vessels in year x is then given by

$$n_2(x) = \frac{K_2}{1 + \exp(-a(x - \tilde{x_0}))}.$$

The result is shown in Figure 5a. Using the information about installed power per ship segment from Table 6, the total number of reactors in the fleet is calculated. This is shown in Figure 5b.



(a) (b)

Figure 5: Baseline deployment of nuclear-powered vessels and SMRs between 2035 and 2060. Panel (a) shows the fleet deployment by ship segment. Panel (b) shows the total number of SMRs per year

The annual sailing distance of the deployed fleet was calculated for each ship segment by combining the number of active vessels in each year with the representative annual distance travelled by one vessel of that segment. For segment i, the annual sailing distance is estimated as

$$D_i(y) = N_i(y)\, v_i\, 1.852\, t_{\mathrm{op},i}, \tag{8}$$

where $N_i(y)$ is the number of deployed vessels of segment i in year y, v_i is the average vessel speed in knots, 1.852 is the conversion factor from knots to km/h, and $t_{\mathrm{op},i}$ is the annual operating time in hours (see Table 6 for average vessel speed and anual operating time). The total annual sailing distance of the fleet is then

$$D_{\mathrm{fleet}}(y) = \sum_i D_i(y). \tag{9}$$

Figure 6 shows that the total sailing activity increases following the same general S-shaped deployment trend as the number of ships in operation (see Figure 5a). However, the contribution of each segment is not determined only by the number of vessels deployed, but also by the average speed and annual operating time assumed for that segment. Large and ultra-large container ships therefore contribute strongly to the total sailing distance because they combine relatively high operating speeds with high annual utilisation. In contrast, other ship segments such as icebreakers and fishing vessels contribute less to the total distance because of their lower representative speeds and smaller fleet shares. This figure provides the maritime activity baseline that will later be used to build the active port-route network.

The annual number of port calls was also calculated for each ship segment. This metric is important because it links the fleet-deployment scenario with the future port-network and service-hub requirements. While the annual sailing distance constrains the amount of route activity that must be assigned to the maritime network, the annual port-call demand constrains how frequently nuclear-powered vessels must be received, inspected, serviced or potentially refuelled at compatible ports.

For each ship segment i, the annual number of port calls is calculated as

$$C_i(y) = N_i(y)\, c_i, \tag{10}$$

where c_i is the assumed number of port calls per vessel per year for that segment (see Table 6 for annual port calls). The total annual number of port calls for the fleet is then

$$C_{\mathrm{fleet}}(y) = \sum_i C_i(y). \tag{11}$$

Figure 7 shows that total port-call demand increases with the fleet deployment, but the relative contribution of each segment differs from the sailing-distance results. In particular, passenger ferries contribute strongly to port-call demand despite representing a relatively small share of the fleet, because they are assumed to make many port calls per year. By contrast, bulk carriers represent a relatively high percentage in the final fleet, but their assumed number of port calls per vessel is comparatively low. This distinction is important for the later port-network model: long-distance route demand is mainly constrained by sailing distance, while the number and distribution of port calls constrains port-service demand.

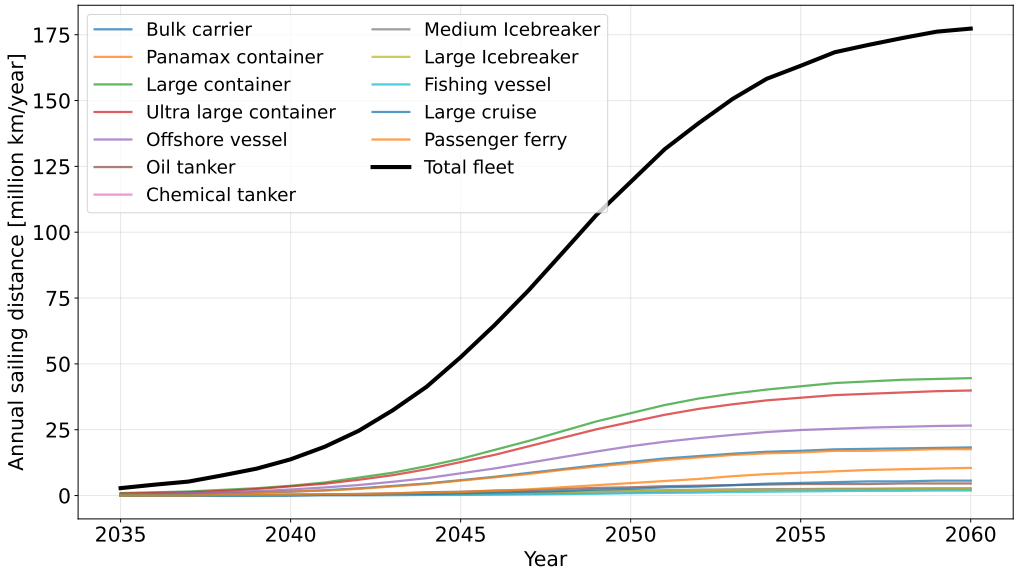


Figure 6: Annual sailing distance by ship segment and total fleet between 2035 and 2060. The black curve shows the total annual sailing distance of the complete nuclear-powered fleet.

5.2 Fleet route-network deployment

The fleet-deployment scenario described above defines how many vessels of each ship segment are active in each year between 2035 and 2060. To translate this fleet deployment into a maritime operating network, two input files are combined. First, `deployment_baseline.txt` specifies the number of active ships by segment and year. Second, `port_connection_proposals.txt` defines the candidate port-to-port connections available to the model. Each route proposal includes the start and end ports, the year from which each port is available in the operating network, an estimated sea-route distance, a route type, a proposed operation year and the ship segments allowed to use the route.

For a route to be available to a given ship segment in year y, three conditions must be satisfied. The route proposed-operation year must be lower than or equal to y, both the start and end ports must already be active in the port network, and the ship segment must be included in the route's list of allowed vessel categories. The route-allocation model is then applied independently to each ship segment. For each year and segment, the model first calculates the annual sailing-distance capacity and the annual port-call demand from the deployed ships. It then assigns route travels to the available candidate routes. One route travel is interpreted as one go-return service, consuming two port calls and a sailing distance equal to twice the one-way route distance.

The allocation algorithm therefore attempts to satisfy two constraints simultaneously: the total sailing distance assigned to routes should be close to the available annual distance capacity of the ship segment, and the number of allocated port calls should match the port-call demand. In practice, exact matching is not always possible because route distances are fixed and route travels are integer values.

The route network is constructed sequentially from 2035 to 2060. This means that, once a route is selected for a given ship segment, it is retained in the following years. This cumulative rule prevents the network from being rebuilt from zero every year and represents a more realistic deployment pathway: early routes remain active, while new regional and intercontinental routes are progressively added as the fleet grows.

Table 11 summarises the route-network deployment for selected years. The fleet grows from 62 ships in 2040 to 800 ships in 2060. Over the same period, annual sailing-distance capacity increases from approximately 13.76 million km/year to 177.31 million km/year, while annual port-call demand increases from 2950 to about 51,500 port calls per year. The active route network also expands, from 10 active routes in 2040 to 34 active routes in 2060.

5.3 From maritime operating ports to nuclear-capable ports

The port network described above represents the ports and routes used by the nuclear-powered fleet for normal maritime operation. The port activation year used in the maritime network should therefore be interpreted only as the year in which the port becomes part of the operating route network. A second classification is needed to describe when a port becomes nuclear-capable.

In the present model, a nuclear-capable port is defined as a port that can support the nuclear fuel-cycle interface with the maritime fleet. This includes fresh-fuel reception, secure fuel storage, reactor refuelling

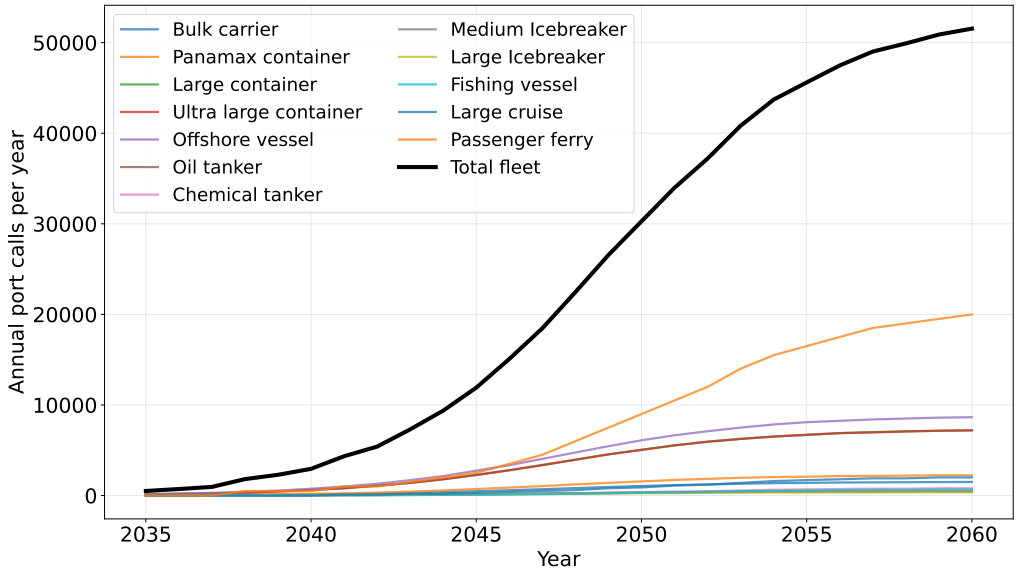


Figure 7: Annual port calls by ship segment and total fleet between 2035 and 2060. The black curve shows the total annual port-call demand of the complete nuclear-powered fleet.

Table 11: Summary of route-network deployment for selected years. Fleet distance and allocated distance are expressed in million km/year.

Year	Ships	Active routes	Fleet distance	Allocated distance	Target port calls	Allocated port calls
2040	62	10	13.76	15.06	2950	2840
2045	237	23	52.54	45.24	11915	11916
2050	537	28	119.08	102.92	30275	30276
2055	736	33	163.20	150.34	45610	45610
2060	800	34	177.31	163.46	51550	51550

operations and interim spent-fuel storage. These activities impose stronger requirements than conventional port operation.

A mature nuclear-powered maritime network cannot rely on only one or two nuclear-capable ports; it requires use of a distributed nuclear-port network in the baseline scenario. For this reason, the model distinguishes between operating ports and nuclear-capable ports. The ports in Table 12 are proposed as nuclear-capable ports and are selected among the ports shown in Table 10.

Table 12: Baseline assumption for the deployment of nuclear-capable ports.

Year	Number of nuclear-capable ports	Ports added to the nuclear-service network
2037	1	Rotterdam
2038	3	Singapore, Hamburg
2040	4	Shanghai
2045	7	Busan, Jebel Ali, Los Angeles
2046	8	Provideniya
2050	10	Algeciras, Murmansk
2055	12	Savannah, Santos
2060	16	Melbourne, Durban, Tromsø, Port Klang

The selected ports are chosen to provide geographical coverage, connection to high-activity routes, and proximity to countries or regions with relevant nuclear, industrial or fuel-cycle capabilities.

The introduction of nuclear-service ports therefore links the maritime route model with the fuel-cycle model. The route network determines where nuclear-powered vessels operate and how often they call at ports, while the fuel-cycle model determines how many refuellings, fresh-fuel deliveries, spent-fuel discharges and cask movements must be supported. The nuclear-port deployment layer connects these two parts of the model by assigning refuelling and waste-management functions to a limited subset of strategically selected ports.

The resulting network expansion is shown in Figures 8–12. The red crosses on the maps indicate the nuclear-capable ports.

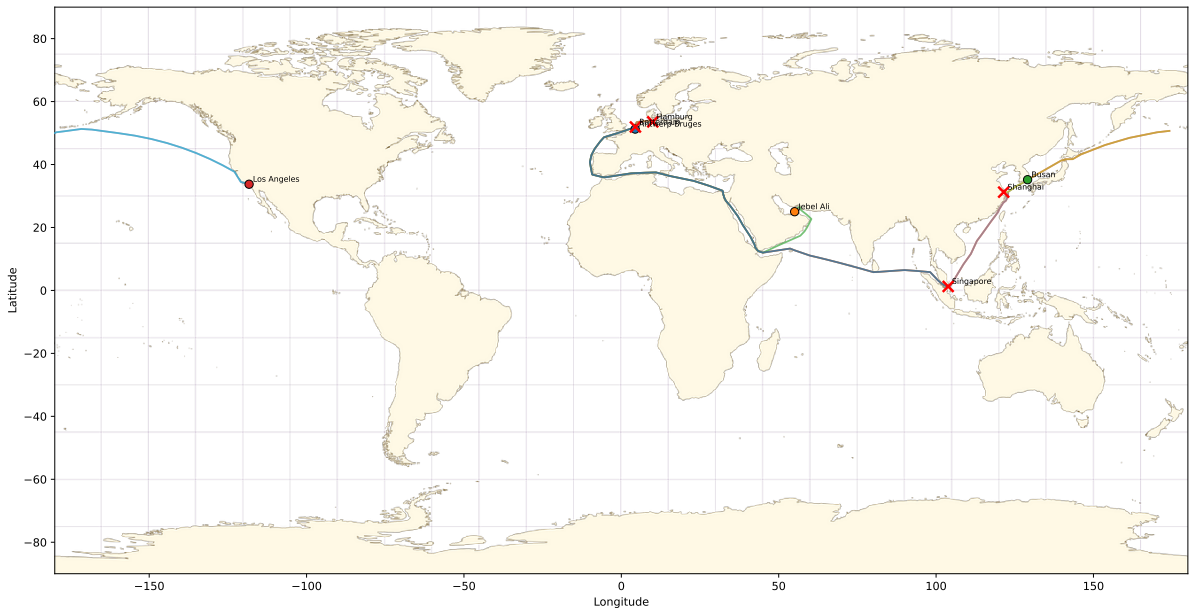


Figure 8: Active route network for all ship segments in 2040.

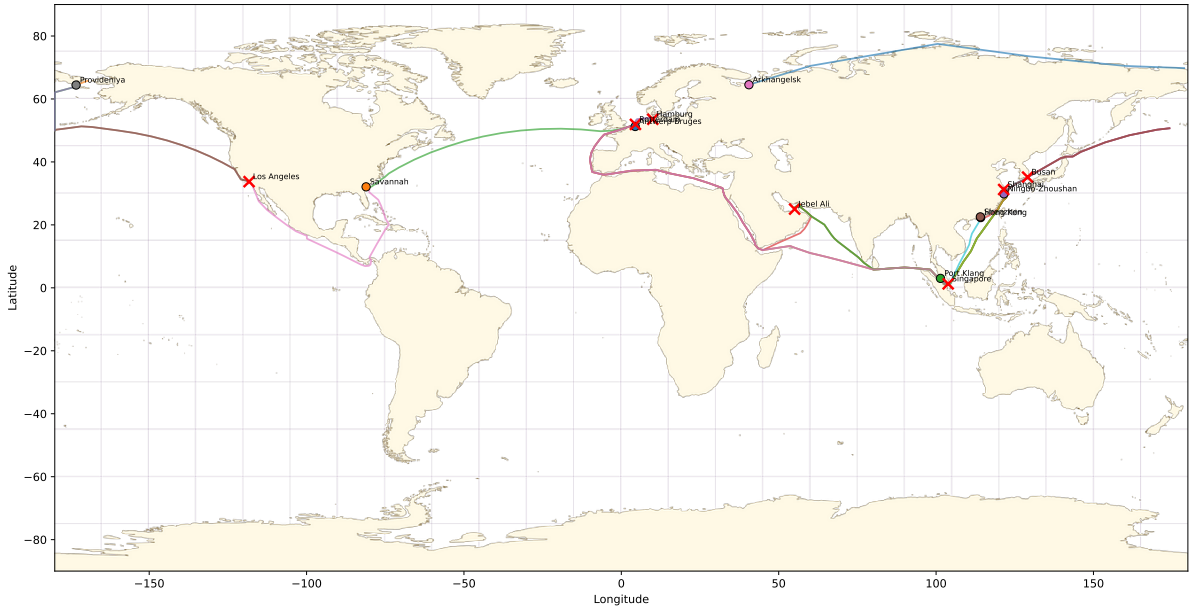


Figure 9: Active route network for all ship segments in 2045.

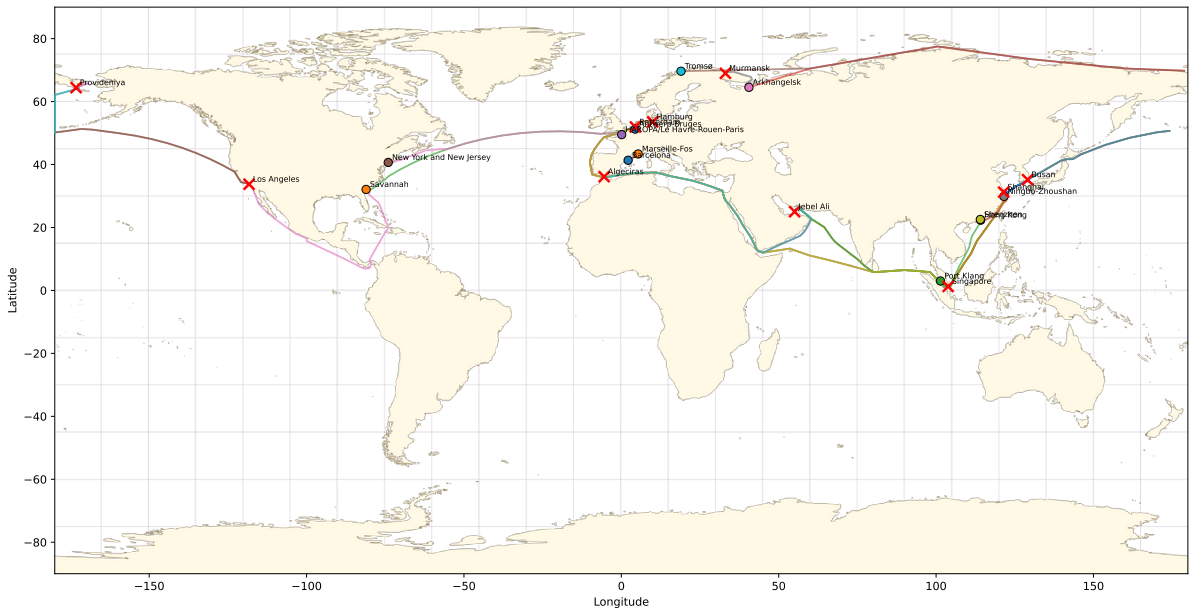


Figure 10: Active route network for all ship segments in 2050.

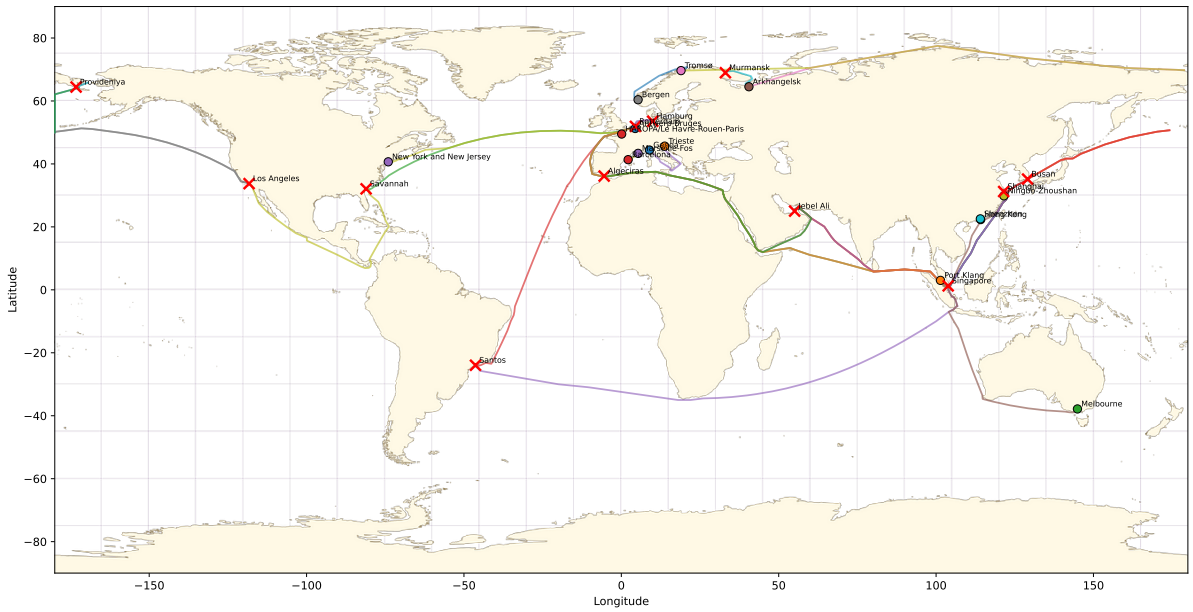


Figure 11: Active route network for all ship segments in 2055.

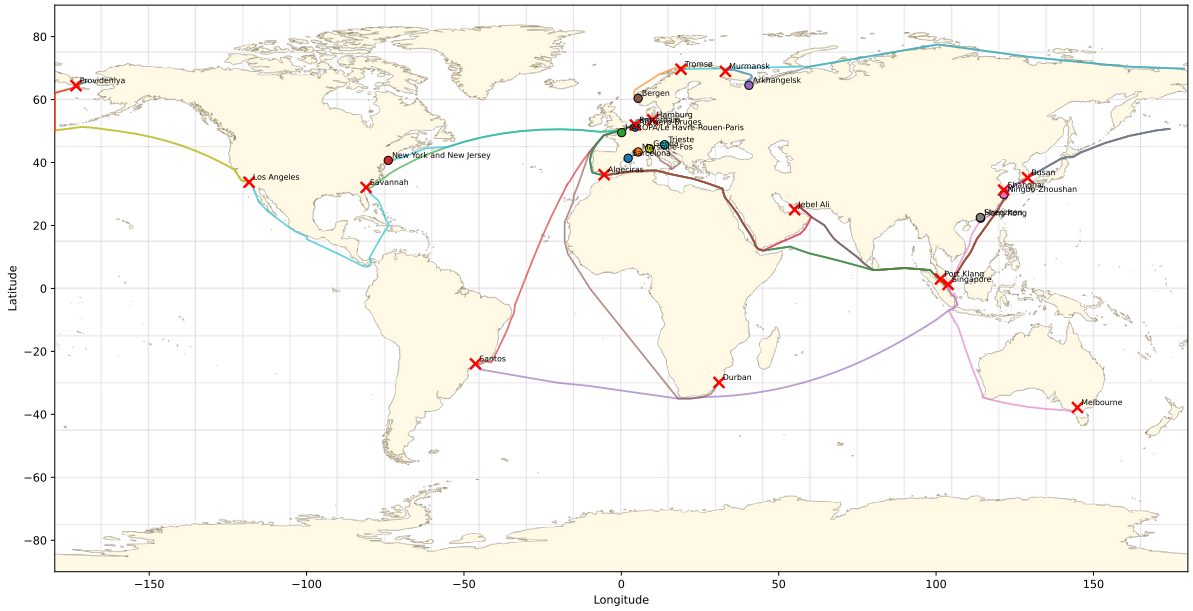


Figure 12: Active route network for all ship segments in 2060.

6 Model calculations and outputs

This section translates the fleet-deployment and route-network assumptions described in Section 5 into quantitative fuel-cycle and infrastructure requirements. The outputs are divided into two groups. First, the fuel-related outputs estimate the fresh-fuel throughput, enrichment effort, refuelling demand and spent-fuel outflow associated with the deployed reactor fleet. Second, the infrastructure-related outputs estimate the order of magnitude of the supply-chain capacity, transport cadence, port-service hub capacity and interim storage required to support the maritime nuclear fleet. These calculations provide the basis for the global and localized results presented in Section 7.

6.1 Fuel related

6.1.1 Fresh fuel throughput

Fresh (uranium) fuel throughput of a single reactor can be calculated using that the reactor type will have an upper limit for burnup,

$$\begin{aligned} \mathrm{BU} &= \frac{P_{\mathrm{reactor,th}} t_{\mathrm{op}} N_{\mathrm{years}}}{m_{\mathrm{fuel}}} \\ \Rightarrow m_{\mathrm{fuel}} &= \frac{P_{\mathrm{reactor,th}} t_{\mathrm{op}} N_{\mathrm{years}}}{\mathrm{BU}}, \end{aligned}$$

where m_{fuel} denotes the total mass of the fuel at beginning of cycle, t_{op} is the operational time per year and N_{years} is the number of years between refuelings. For a fleet consisting of N different ship segment, with $n_{\mathrm{reactor},i}$ reactors per ship for segment i, $n_{\mathrm{ship},i}$ number of ships of segment i, where each reactor has a fuel throughput of $m_{\mathrm{fuel},i}$, the total fuel throughput is given by

$$\sum_{i=1}^{N} n_{\mathrm{reactor},i} \cdot n_{\mathrm{ship},i} \cdot m_{\mathrm{fuel},i}.$$

Having found the total uranium throughput of the fleet, this can be translated to the required TRISO mass. Of a 200 g TRISO particle, around $7-9$ g will be uranium [27]. Based on this, assume that the uranium mass constitutes 4% of the total TRISO mass.

$$m_{\mathrm{TRISO}} = \frac{m_{\mathrm{fuel}}}{0.04}$$

The number of TRISO pebbles can be similarly found by dividing m_{fuel} by the uranium mass per pebble, $m_{\mathrm{U,pebble}} \sim 8$ g.

6.1.2 Enrichment effort

The mass balance over an enrichment cascade dictates that all of the material that is fed into the plant appears in the tails and product, provided losses can be ignored. This leads to a basic equation: $P = F - W$, where P is the product mass, F is the feed mass and W is the tail mass [28]. The separative work needed to produce P mass of uranium enriched to a mass assay of x_p is given by

$$\mathrm{SW} = P(1-2x_p)\ln\frac{1-x_p}{x_p} + W(1-2x_w)\ln\frac{1-x_w}{x_w} - F(1-2x_f)\ln\frac{1-x_f}{x_f},$$

where F is the feed mass, x_f is the assay of the feed mass, W is the mass of the tails, and x_w is the assay of the tails. For given assays of the product, feed and tail masses, one can calculate the feed mass needed to produce a product mass P,

$$F = P\frac{x_p - x_w}{x_f - x_w}.$$

Higher tail assay will require less separative work and more feed mass, and vice versa for lower tail assay. As an assumption, natural uranium is used as feed, with a tail assay of 0.40%.

6.1.3 Refueling requirements

Given that there are N_{years} years between refuelings, each reactor would need to be refueled with

$$n_{\text{TRISO,pebble}} = \frac{m_{\text{fuel}}}{m_{\text{U,pebble}}} \tag{12}$$

pebbles, where m_{fuel} is the fuel mass discussed above and $m_{\text{U,pebble}}$ is the uranium mass content of each pebble. The number of ships which would need refueling in a given year, is simply the number of ships fueled and refueled N_{years} years prior. The calculation follows a similar logic as in Algorithm 1.

6.1.4 Spent fuel outflows

The spent fuel will be encapsulated in the TRISO pebbles. Several methods exist for removing the coatings, e.g. acid leaching, high voltage pulse fragmentation and grinding, which could lower the disposed volume by up to 90% [2]. Relevant to storage is the volume and packing fraction of the disposed pebbles. The number of disposed pebbles is equal to the number of pebbles loaded in the reactor at the previous refueling, $n_{\text{TRISO,pebble}}$. From there, the total volume can be calculated as

$$V_{\text{waste}} = n_{\text{TRISO,pebble}} V_{\text{TRISO,pebble}} \tag{13}$$

$$= n_{\text{TRISO,pebble}} \frac{\pi}{6} d^3_{\text{pebble}}, \tag{14}$$

where d_{pebble} is the diameter of the pebble, typically around 6 cm [27].

6.2 Infrastructure related

6.2.1 Port-service hub allocation method

The global fuel-cycle and infrastructure calculations described above provide fleet-wide requirements for each year, such as refuelling, fresh-fuel storage, spent-fuel storage and spent-fuel transport. These quantities are global model outputs: they describe the total requirement of the complete nuclear-powered fleet. However, these global requirements need to be distributed among the nuclear-capable ports introduced in Table 12. The purpose of this subsection is to define the allocation method used to assign that demand to individual ports.

The model defines the set of active nuclear-capable ports in year y as

$$P_{\text{nuc}}(y) = \{p : y \geq y_{\text{nuc},p}\},$$

where $y_{\text{nuc},p}$ is the nuclear-capability activation year of port p. A port can only receive fuel-cycle responsibilities after this activation year. The activation year should not be interpreted as a statement of current regulatory readiness, but as a scenario assumption for when the port has developed the required nuclear-service infrastructure.

For a generic global infrastructure requirement $X_{\text{global}}(y)$, the requirement allocated to port p is written as

$$X_p(y) = s_p(y) X_{\text{global}}(y),$$

where $s_p(y)$ is the allocation share of port p in year y. The allocation share is calculated as

$$s_p(y) = \frac{w_p f_p(y)}{\sum_{q \in P_{\text{nuc}}(y)} w_q f_q(y)}.$$

Here, w_p is the nominal port weight and $f_p(y)$ is the maturity factor. The nominal weight w_p represents the relative share of the global service burden that a port is assumed to take once mature. A port with a larger w_p receives a larger fraction of the global requirement, while a port with a smaller w_p receives a smaller fraction.

Because the allocation shares are normalized by the sum of all active weighted ports, the total allocated requirement is conserved:

$$\sum_{p \in P_{\text{nuc}}(y)} X_p(y) = X_{\text{global}}(y) \sum_{p \in P_{\text{nuc}}(y)} s_p(y) = X_{\text{global}}(y).$$

Thus, the port weights do not increase or decrease the global fuel-cycle requirement. They only determine how that fixed global requirement is distributed among the active nuclear-capable ports.

A maturity factor is introduced because a newly activated nuclear-capable port is unlikely to operate immediately at full service-hub capacity. In the baseline case, a port is assigned 50% of its nominal weight in its activation year, 75% in the following year, and 100% from the third year onward. Therefore, a newly activated port initially receives a smaller share of the global requirement than it would receive once fully mature.

In this version of the model, the port weights w_p are made port-specific in order to reflect two first-order effects: proximity to assumed HALEU/TRISO fuel-cycle regions (read subsection 3.2) and maritime centrality in the route-allocation model (see Figures 8–12).

Based on these criteria, the following nominal port weights are used in the baseline service-hub allocation:

$$\begin{aligned}
&w_{\text{Rotterdam}} = 1.30, \quad w_{\text{Singapore}} = 1.35, \quad w_{\text{Hamburg}} = 1.30, \quad w_{\text{Shanghai}} = 1.50,\\
&w_{\text{Provideniya}} = 0.80, \quad w_{\text{Busan}} = 1.15, \quad w_{\text{JebelAli}} = 1.40, \quad w_{\text{LosAngeles}} = 0.95,\\
&w_{\text{Algeciras}} = 1.50, \quad w_{\text{Murmansk}} = 1.15, \quad w_{\text{Savannah}} = 1.25, \quad w_{\text{Santos}} = 0.90,\\
&w_{\text{Melbourne}} = 0.70, \quad w_{\text{Durban}} = 0.85, \quad w_{\text{Tromso}} = 1.00, \quad w_{\text{PortKlang}} = 1.20.
\end{aligned}$$

The highest weights are assigned to Shanghai and Algeciras. Shanghai is favoured because it is both a major East Asian route node and located in a country assumed to have domestic HALEU and TRISO-related capability. Algeciras is favoured because it is one of the most central ports in the allocated route network and provides a strategic connection between the Mediterranean, Suez corridor and Atlantic routes. Jebel Ali, Singapore, Rotterdam and Hamburg are also assigned relatively high weights because they are major logistics hubs in the operating network and are either close to fuel-cycle regions or highly central in the modelled route structure.

These weights remain simplified scenario assumptions. A complete port-capacity assessment would require a dedicated study of licensing, security, land availability, national regulatory frameworks and more. This regulatory and infrastructure-readiness timescale is not studied in detail in the present project. It is therefore identified as an important direction for further development of the route-network and nuclear-port model.

6.2.2 Interim storage needs

Ports with refueling capabilities must include interim storage capacity for both fresh and spent fuel.
Start by defining the global total storage capacity needed to supply the fleet. Assuming a N_{year} year refueling interval, the storage need for year $x \geq N_{\text{year}}$ after 2035 must be the sum of the fuel loaded into the ships deployed in year x and the refueling of the ships from year $x - N_{\text{year}}$. The storage need for a given year is then calculated sequentially as described in Algorithm 1.

Algorithm 1 Fresh fuel storage needed per year

Require: year ≥ 2035
  storage $\leftarrow$ [fuelneed $\cdot N_{\text{ships}}$[year = 2035]]
  for year $\leq$ end_year **do**
    if year$-N_{\text{years}} \geq 0$ **then**
      refuel $\leftarrow$ storage[year$-N_{\text{years}}$]
    else
      refuel $\leftarrow$0
    end if
    storage[year] $\leftarrow$ fuelneed $\cdot(N_{\text{ships}}[\text{year}] - N_{\text{ships}}[\text{year} - 1])$+refuel
  end for

The rise in the spent fuel storage need will be delayed by N_{years} years compared with the rise in fresh fuel storage need. This is because the spent fuel handling only becomes relevant for refueling, not for the initial deployment of a ship. The calculation of the spent fuel storage need is coupled to the fresh fuel storage need. This is described by Algorithm 2.

Algorithm 2 returns the total volume of the fresh and spent TRISO pebbles being stored in the ports. However,

Algorithm 2 Fresh and spent fuel storage needed per year

Require: year ≥ 2035
  fresh_store $\leftarrow$ [fuelneed $\cdot N_{\text{ships}}$[year = 2035]]
  for year $\leq$ end_year **do**
    if year$-N_{\text{years}} \geq 0$ **then**
      refuel $\leftarrow$ fresh_store[year$-N_{\text{years}}$]
      spent_store[year] $\leftarrow$ spent_store[year$-$1]+fresh_store[year$-N_{\text{years}}$]
      if year$-N_{\text{years}}$ $-$ store_years ≥ 0 **then**
        spent_store[year] $\leftarrow$ spent_store[year]-fresh_store[year$-N_{\text{years}}-$store_years]
      end if
    else
      refuel $\leftarrow$0
      spent_store[year] $\leftarrow$ 0
    end if
    storage[year] $\leftarrow$ fuelneed $\cdot(N_{\text{ships}}[\text{year}] - N_{\text{ships}}[\text{year} - 1])$+refuel
  end for

these pebbles will naturally be stored in some type of container, increasing the total volume of the storage space. Two previous experience with spent fuel storage are releated to the AVR and HTR-10 pebble bed reactors.

The AVR research reactor in Germany stored the spent fuel in CASTOR casks, in dry storage [29, pp. 573-574]. Details are shown in Table 13.

Outer/inner length [mm]	2780/1964
Outer/inner diameter [mm]	1380/640
Weight [t]	26.1
Number of pebbles	1900
Pebble packing fraction [%]	34

Table 13: Dimensions of CASTOR THTR/AVR cask [29, p. 580]. The final row was calculated assuming 60 mm diameter pebbles filling the interior of the cask.

The spent fuel from the HTR-10 research reactor in China was similarly in dry storage, in stainless steel containers, cooled by natural air convection [29, p.606]. Its dimensions are shown in Table 14.

Outer height [mm]	1000
Outer diameter [mm]	622
Number of pebbles	1000

Table 14: Dimensions of the spent fuel storage containers at the HTR-10 facility [29, p. 606].

However, to estimate future storage requirements, it is useful to compare with newer packaging concepts like the Optimus casks provided by NAC International [30]. The Optimus casks come in to variants, the lightweight Optimus-L, and the Optimus-H with increased radiation shielding. The former has been approved for transport of fresh HALEU TRISO fuel [31] by the US Nuclear Regulatory Commission. For the purpose of this analysis, it is in the following assumed that the Optimus-H cask will receive similar approval for dry storage and transport of spent TRISO fuel by 2035. The additional radiation shielding in the Optimus-H cask is necessary to deal with the radioactivity of fission products in the spent fuel, though this shielding comes at the expense of higher weight and a smaller cavity volume. The dimensions of both casks are shown in Table 15.

	Optimus-H	Optimus-L
Outer/inner height [cm]	211.3/119.4	177.8/115.6
Outer/inner diameter [cm]	188.5/82.6	124.5/82.6

Table 15: Dimensions of Optimus-H and Optimus-L casks [30].

The total uranium mass loaded per Optimus-L cask must be ≤ 68 kg [32]. Furthermore, geometry places a limit on how many TRISO pebbles can physically fit inside the cask cavity volume. For both casks, the cavity diameter is ~ 14 times larger than the pebble diameter. Finding the optimal packing of spheres in a cylinder for this configuration is outside the scope of this analysis. For simplicity, assume at most half of the cavity volume is filled by TRISO pebbles on average.
With $m_{\mathrm{U,cyl}} \leq 68$ kg and $m_{\mathrm{U,pebble}} = 8$ g, at most $N_{\mathrm{pebbles,cyl}} = 68 \cdot 10^3/8 = 8500$ pebbles can fit in the Optimus-L cask. However, since $V_{\mathrm{pebble}} N_{\mathrm{pebble,cyl}} \leq V_{\mathrm{cavity}}/2$

$$\Rightarrow N_{\mathrm{pebble,cyl}} \leq \frac{V_{\mathrm{cavity}}}{2V_{\mathrm{pebble}}} = \frac{1}{2}\frac{\pi/4}{\pi/6}\frac{d^2_{\mathrm{cavity}} h_{\mathrm{cavity}}}{d^3_{\mathrm{pebble}}} \tag{15}$$

$$= \frac{3}{4}\frac{82.6^2 \cdot 115.6 \text{ cm}^3}{6^3 \text{ cm}^3} \approx 2739 \tag{16}$$

Therefore, only $N_{\mathrm{pebble,cyl}} = 2739$ pebbles can fit in each Optimus-L cask, well below the uranium weight limit.

For the Optimus-H cask, the same assumptions regarding sphere packing are assumed. Using Equation 15, the maximum number of pebbles is found to be $N_{\mathrm{pebble,cyl}} = 2829$.

Using the values in Tables 13, 14 and 15, the required cylinder volume per pebble can be calculated and compared. The results of this calculation are shown in Table 16.

	AVR	HTR-10	Optimus-H	Optimus-L
Volume per pebble [m^3]	$2.19 \cdot 10^{-3}$	$3.04 \cdot 10^{-4}$	$2.08 \cdot 10^{-3}$	$7.90 \cdot 10^{-4}$

Table 16: Outer cylinder volume required per stored TRISO pebble.

One observes from Table 16 that the Optimus casks provide a storage volume per TRISO pebble comparable to that used for the AVR and HTR-10 reactors, although the casks used for the HTR-10 are the most compact. This analysis nevertheless chooses the Optimus casks as a basis, since these can also be used for transport [30]. Then the required storage volume per stored fresh pebble is ~ 18.43265 times larger than the volume of a 6 cm diameter sphere, while it is ~ 6.9885 larger for a spent pebble. Arranging the cylinders in a square grid, one can at most cover $\pi/4$ of the floor area. The actual required storage space must then be multiplied by a factor $4/\pi$, so that the required volume to store a pebble is ~ 25 times its own volume. Based on this, a conversion factor $f_{\mathrm{geo,fresh}} = 6.9885 \cdot 4/\pi$ is used to convert pebble volume to required storage volume the fresh fuel, while $f_{\mathrm{geo,spent}} = 18.43265 \cdot 4/\pi$ is used for the spent fuel. Dividing by the respective heights of the casks yields the required floor space. One can assume that the spent fuel will have to be stored for at least $4 - 5$ years before being moved away from the port facility, given thar the TN-FSV cask previously used for transport in the US required 1600 days of cooling [33].

6.2.3 Transport and security cadence

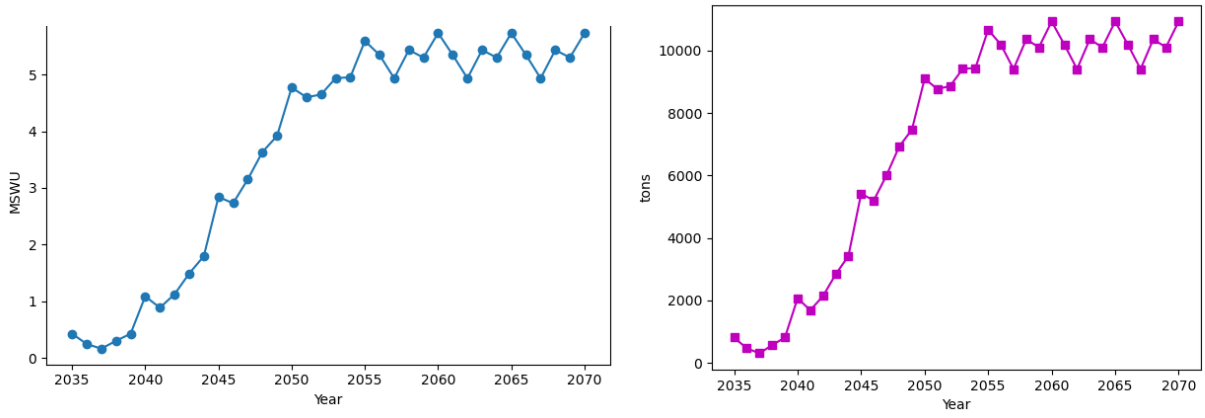
HALEU fuel can only be handled by Category II facilities in the US [12], and corresponding safety regulation exists for other countries as well. Every facility which handles nuclear material requires security infrastructure and personnel. Transport of nuclear material is under a similarly strict regime. Fuel must be transported to ports to meet refueling demands, while spent fuel must be transported away from the port at some frequency.

While outside the scope of this model, there must be infrastructure for processing, long-term storage and eventual deposition of the spent nuclear fuel. For the purposes of this analysis, the details of what happens after the spent fuel is moved out of interim storage at the ports are neglected.

Previous transport efforts include transport from Fort St. Vrain using the TN-FSV cask [33]. One flatbed truck was required per cask in that case. For the Optimus-H cask, however, two casks can be transported on a single truck [34]. The number of truckloads needed to transport the spent fuel in a year is then simply found by dividing the total volume of the casks removed that year by $2 \cdot V_{\mathrm{cyl}}$, where V_{cyl} is the outer volume of the Optimus-H cask.

7 Results

The results are divided into two sections, the global fuel and infrastructure requirements of the fleet, and those needs broken down to individual countries and ports.

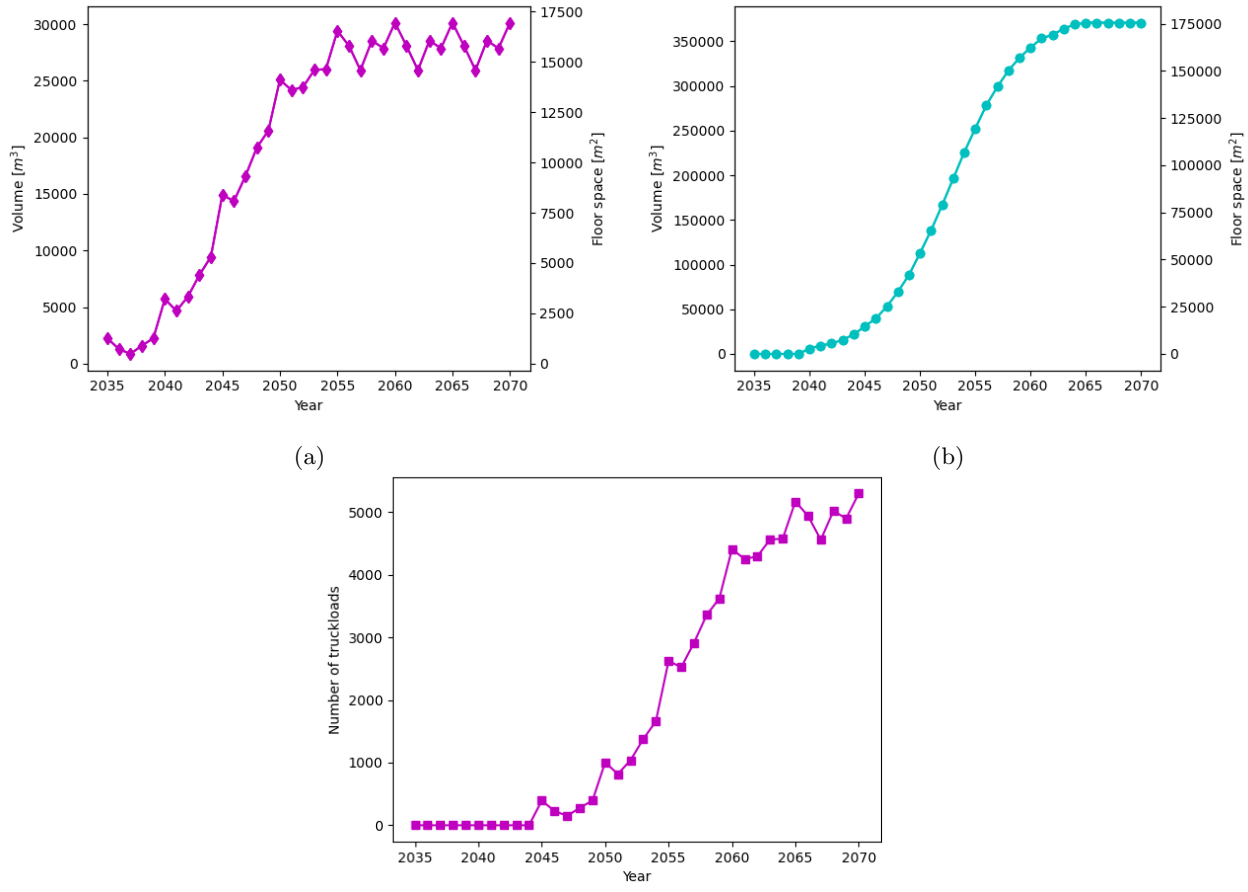


(a) (b)

Figure 13: Annual global enrichment requirements for the baseline fleet scenario. Panel (a) shows the separative work required to produce the HALEU fuel used by the fleet, expressed in MSWU/year, with 0.4% tail assay. Panel (b) shows the corresponding annual natural-uranium feed requirement for the enrichment step, expressed in tonnes of uranium per year, with 0.4% tail assay.

7.1 Global

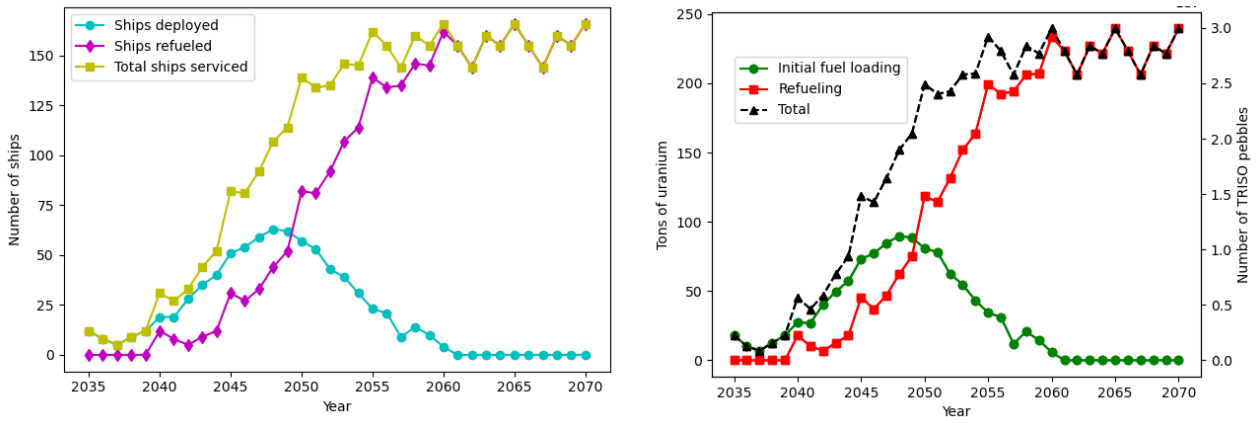
The global model outputs are shown in Figures 13, 14 and 15.



(a) (b)

(c)

Figure 14: Global port-infrastructure requirements in the baseline scenario considering 5 years refueling interval and 5 years of spent fuel storage. Panel (a) shows the yearly fresh-fuel storage capacity required at ports to support fuel loading. Panel (b) shows the yearly spent-fuel interim storage needs. Panel (c) shows the annual number of spent-fuel truckloads required to remove spent fuel after the assumed storage period.



(a) (b)

Figure 15: Ship deployment and refueling requirements. Panel (a) shows the number of service events per year. Panel (b) shows the yearly fuel throughput and refueling requirements.

7.1.1 Order of magnitude requirements for enrichment and fuel fabrication capacity

Production of TRISO fuel is one of the main bottlenecks from a supply chain perspective. From Figure 15b, it can be seen that after 2060, the yearly requirement for fuel is ~ 30 million TRISO pebbles. To produce the pebbles, ~ 240 tons of HALEU must be produced. Table 17 shows the required fuel for selected years.

Year	2040	2045	2050	2060
TRISO [tU]	45.4	118.7	199.5	239.7

Table 17: TRISO fuel requirements for different years of the fleet deployment, in units of tons of enriched uranium.

Comparing Tables 8 and 17, one can see that the TRISO fuel required by the fleet in 2040 corresponds to $\sim 226\%$ of the global production in 2035, $\sim 151\%$ if one includes the Japanese production in 2040. Between 2040 and 2060, the yearly fuel requirements of the fleet increase by a factor ~ 5. In order to supply the fleet, the global production of TRISO would need to increase by an order of magnitude compared to the projections summarized in Table 8. Similarly, production of HALEU would need to increase by a factor ~ 3. If more onshore pebble bed HTGRs are constructed before 2060, the global production would need to increase further.

Another bottleneck is the HALEU enrichment capacity. From Figure 13a, it can be seen that the global enrichment requirements for the fleet in 2060 is ~ 5.7 MSWU.

From Figure 13b, it can be seen that the yearly natural uranium requirement is ~ 10000 tons in 2060. Comparing with Figure 3, this corresponds to $\sim 1/6$ of the global annual production in 2024 and a similar share of the global uranium hexafluoride conversion capacity, as seen from Figure 4.

7.2 Localized

The global fuel-cycle and infrastructure requirements presented in Section 7.1 describe the total demand of the complete nuclear-powered fleet. However, these requirements must ultimately be handled by a limited number of nuclear-capable ports. The purpose of the localized analysis is therefore to translate the global fuel-loading, refuelling, storage and spent-fuel transport requirements into port-level service-hub responsibilities.

Table 18 summarizes the localized service-hub requirements for selected years. Refuelling is already present in 2040: the first vessels deployed in 2035 reach the end of the assumed five-year refuelling interval, resulting in 12 refuellings in addition to 19 initial fuel loadings. By 2060, when the fleet is close to saturation, only four initial fuel loadings are required, while 162 refuellings account for almost all of the 166 annual service events. Over the same period, the number of active nuclear-capable ports increases from four to 16, distributing the growing service burden across a broader network.

Figure 16 shows the average service burden per active nuclear-capable port. The average number of total service events rises during the main fleet-deployment phase and reaches a maximum of approximately 14.6 events per active port in 2053. The refuelling component continues to increase for several more years, reaching approximately 12.2 refuellings per active port in 2058. The decrease visible in 2060 is caused mainly by the expansion of the nuclear-service network from 12 to 16 active ports rather than by a reduction in global refuelling demand. In fact, global refuellings increase from 139 in 2055 to 162 in 2060.

Table 18: Localized nuclear service-hub requirements for selected years. Service events include initial fuel loading and refuelling.

	2040	**2045**	**2050**	**2055**	**2060**
Active ports	4	7	10	12	16
Initial fuel loadings	19	51	57	23	4
Refuellings	12	31	82	139	162
Service events	31	82	139	162	166
Mean service events/port	7.8	11.7	13.9	13.5	10.4
Spent fuel storage [m^3]	5921.7	30825.9	112813.2	252468.3	343008.3
Mean spent fuel storage/port [m^3]	1480.4	4403.7	11281.3	21039.0	21438.0
Fresh fuel storage [m^3]	5713.5	14935.0	25095.0	29418.0	30147.1
Mean fresh fuel storage/port [m^3]	1428.4	2133.6	2509.5	2451.5	1884.2
Total truckloads	0	395	1004	2624	4408
Mean truckloads/port	0.0	56.4	100.4	218.7	275.5

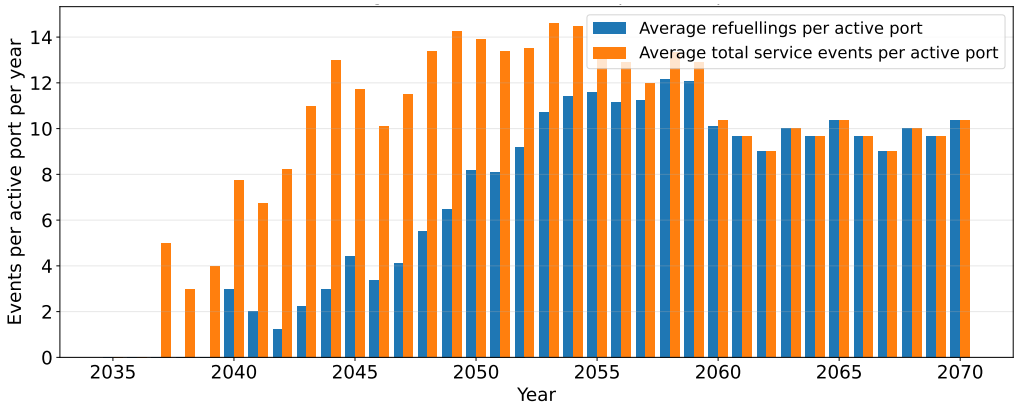


Figure 16: Average annual nuclear service events and refuellings per active nuclear-capable port. Service events include both initial fuel loading for newly deployed ships and refuelling of previously deployed ships.

Figure 17 shows the corresponding average fresh-fuel and spent-fuel storage burden. Average fresh-fuel storage increases during the deployment phase and reaches a maximum of approximately 2.6×10^3 m^3 per active port around 2054. It then decreases as fleet deployment slows and the number of nuclear-capable ports increases. Spent-fuel storage follows a delayed and much stronger increase because spent fuel is generated only after refuelling and remains in interim storage before removal. The average spent-fuel storage burden reaches approximately 2.76×10^4 m^3 per active port in 2059, before falling to about 2.14×10^4 m^3 in 2060 when four additional nuclear-capable ports enter the network.

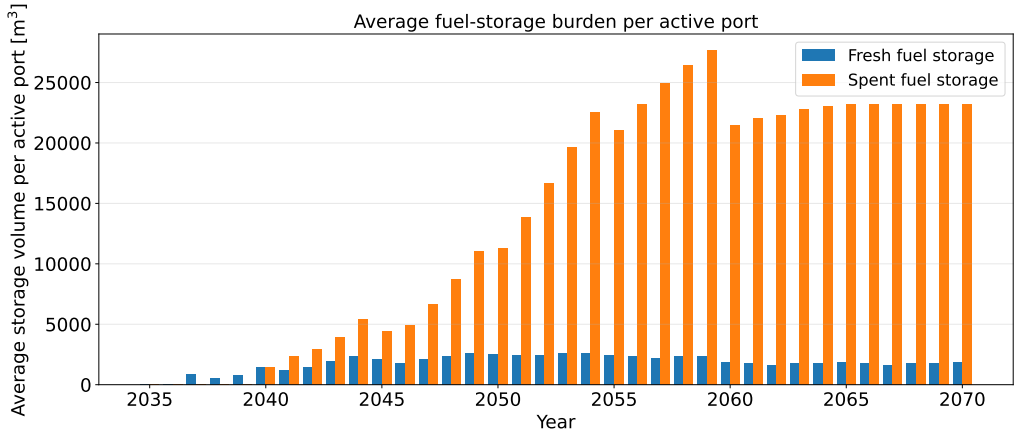


Figure 17: Average fresh-fuel and spent-fuel storage burden per active nuclear-capable port.

Figures 18–20 show how the localized requirements are distributed among individual nuclear-capable ports in

selected years. In 2060, Shanghai and Algeciras receive the largest allocations. Each is assigned approximately 14.8 refuellings per year, about 3.41×10^4 m^3 of total fresh- and spent-fuel storage, and approximately 403 spent-fuel truckloads per year. Jebel Ali and Singapore follow with approximately 13.8 and 13.3 refuellings per year, respectively, while Rotterdam and Hamburg each receive approximately 12.8 refuellings per year. This concentration reflects the higher nominal service weights assigned to the most central ports and to ports located close to assumed fuel-cycle regions.

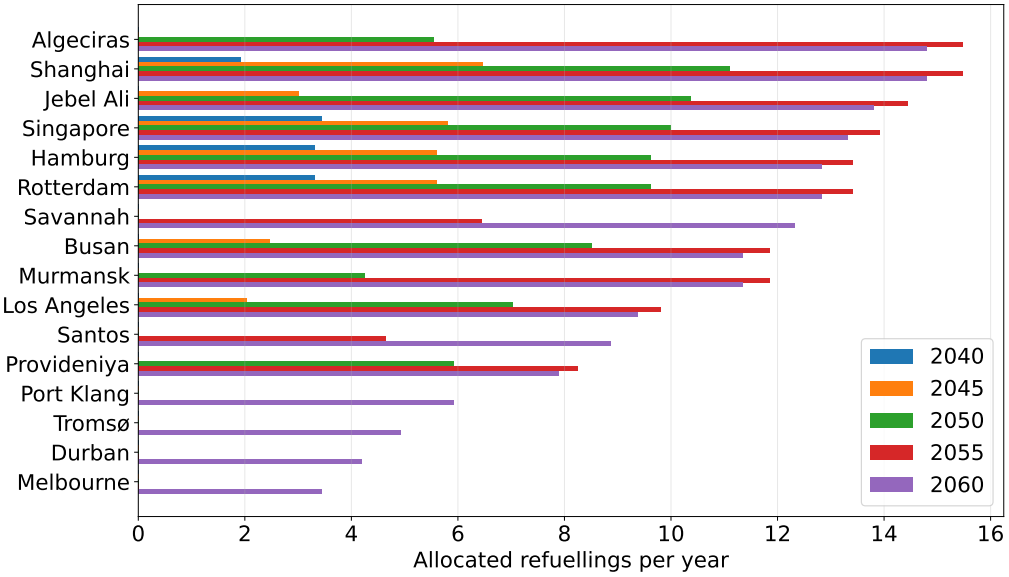


Figure 18: Allocated annual refuellings by nuclear-capable port for selected years.

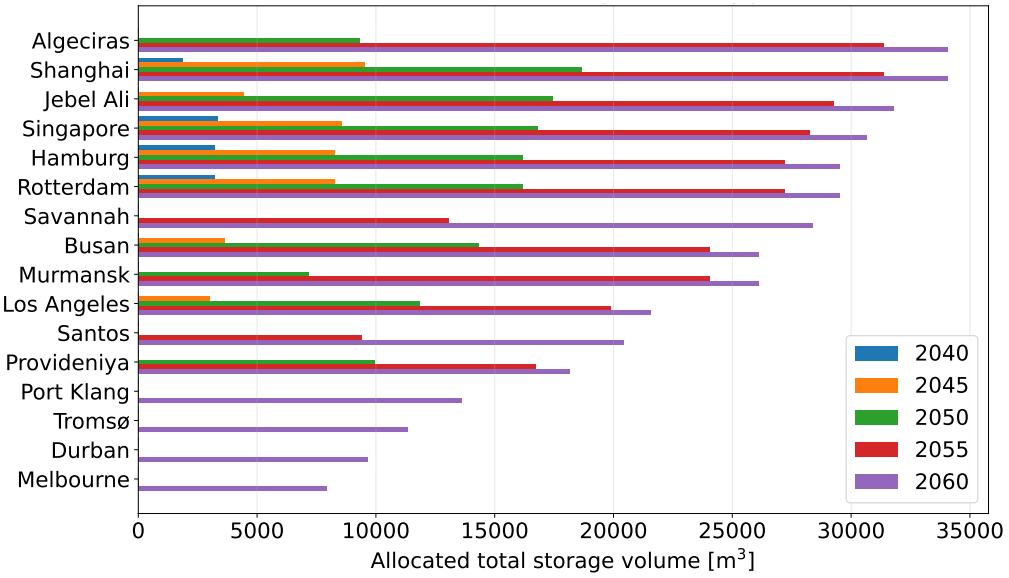


Figure 19: Allocated total fuel-storage volume by nuclear-capable port for selected years. Total storage includes both fresh-fuel and spent-fuel storage.

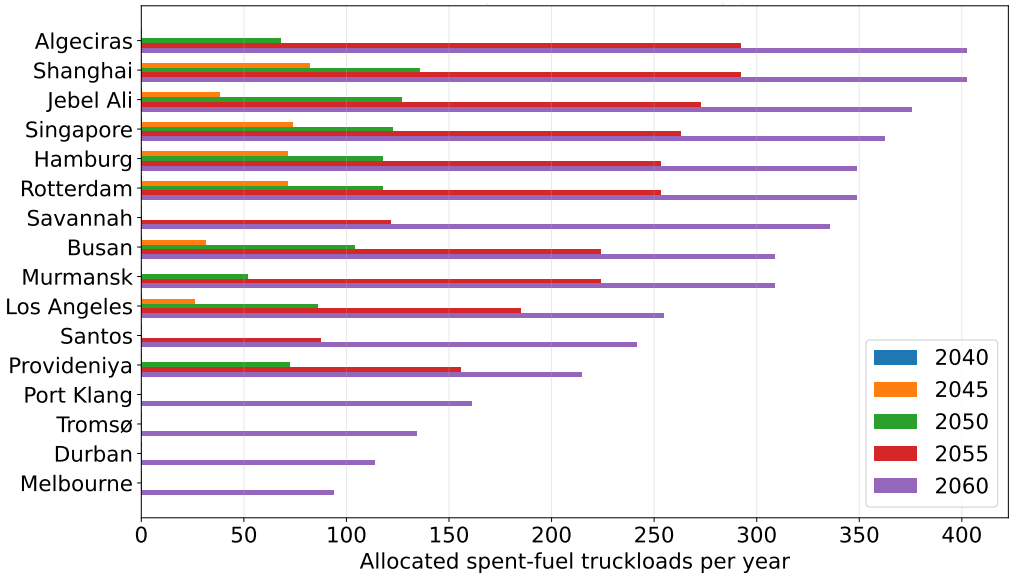


Figure 20: Allocated spent-fuel truckloads by nuclear-capable port for selected years.

The port-level truckload results should be interpreted as allocated transport-handling burdens rather than as a detailed tracing of individual spent-fuel batches. The model first calculates the global spent-fuel outflow after the assumed interim-storage period and then distributes that global requirement among the nuclear-capable ports according to the annual allocation shares. Consequently, a truckload assigned to a given port in a particular year is not necessarily associated with spent fuel that was discharged at that same port exactly five years earlier. A port-specific fuel-history and inter-port logistics model would be required to resolve that level of detail.

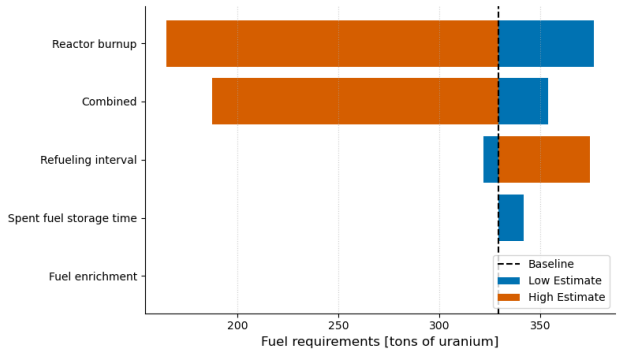
8 Sensitivity analysis

As mentioned in Section 2, assumptions were made regarding the unknown reactor characteristics, specifically burnup, enrichment and refueling intervals. Furthermore, there is uncertainty regarding how long spent fuel should cool in dry storage in the ports before being transported to longer term storage facilities. A sensitivity analysis was performed to investigate how these inputs affect critical global model outputs, in particular the separative work needed, the fuel requirements, the storage needs and number of truckloads of spent fuel being transported. Additionally, the combined effects of varying enrichment, burnup and refueling intervals simultaneously were studied, to account for the high degree of correlation between these quantities in real life. The high, low and baseline estimates for the inputs are summarized in Table 19. The corresponding outputs are illustrated by Figure 21, which shows the maximum value for each output in each scenario. Note that the high and low estimates are not symmetric around the baseline estimate. For instance, the baseline estimate of burnup is on the lower end of what pebble-bed HTGRs can achieve, while the high estimate for burnup is twice the value of the baseline. For the fuel enrichment, on the other hand, the high and low estimates are almost symmetrical around the baseline. The baseline estimates were deliberately chosen to be conservative, so that for most of the inputs, the baseline estimate is closer to the lower estimate than the higher one. However, one should not that the relatively low estimate for the spent fuel storage time is not a strictly conservative estimate. This baseline input value was chosen to match the refueling interval. Since the refueling requirements increase substantially after 2050, see Figure 15b, a spent fuel storage time longer than ~ 5 years will result in the storage needs peaking after 2060. The simulation is run for an additional refueling interval + storage time years after the ship deployment is finished to account for this.

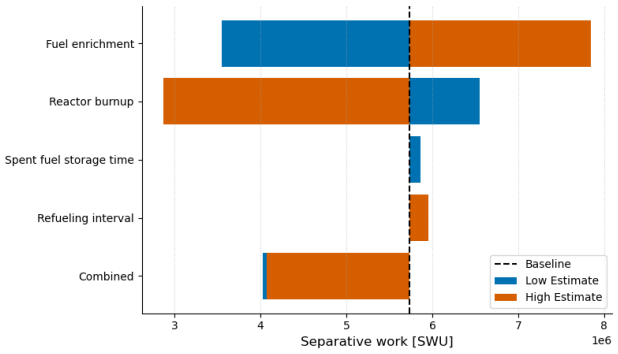
	Low	Baseline	High
Burnup [GWd/t]	70	80	160
Enrichment [%]	10	15	19.75
Refueling interval [years]	4	5	7
Spent fuel storage time [years]	4	5	10

Table 19: High, low and baseline estimates of model inputs.

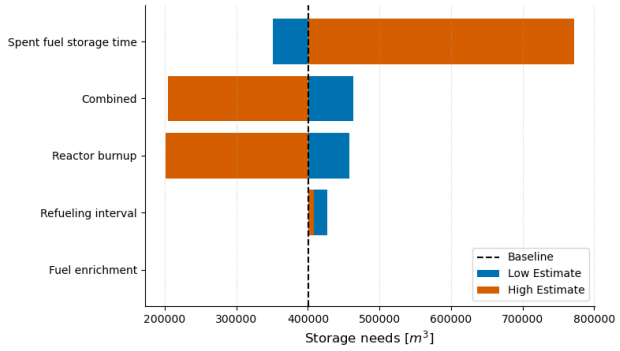
From Figure 21a, it can be seen that the peak yearly fuel requirements of the fleet are only affected by



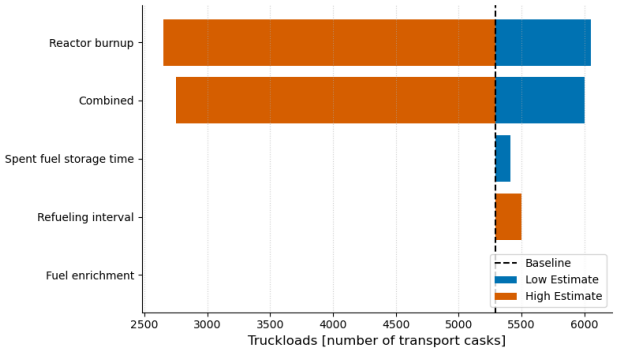
(a) Peak global fuel requirements of fleet. Sum of initial fuel loadings and refulings.



(b) Peak global seperative work needed to supply fleet.



(c) Peak global fuel storage needs. Sum of fresh and spent fuel storage.



(d) Peak global truckloads needed to empty interim spent fuel storage facilities.

Figure 21

the burnup and refueling interval, according to this simplified model. In reality, fuel enrichment is tied to both burnup and time between refuelings. In this model, however, they are treated as independent variables. Similarly, the model treats reactor burnup and refueling intervals as independent variables as well. The estimate of the combined effects of varying burnup, refueling interval and enrichment shows that the high estimate scenario requires less fuel than the low estimate scenario. What Figure 21a does illustrate is that higher fuel burnup is associated with lower fuel requirements, because more energy is extracted per ton of uranium, and vice versa. The figure also shows that more frequent refuelings reduce the fuel requirements, because less fuel is loaded on each ship.

The sensitivity analysis of the required separative work is summarized in Figure 21b. It can be seen that, naturally, the required separative work increases with higher fuel enrichment. The required separative work to produce a certain amount of enriched uranium is generally not a linear function of the enrichment level. Indeed, a near doubling of the fuel enrichment, from 10% to 19.75%, leads to more than a doubling of the required separative work, from 3.6 MSWU to 7.8 MSWU. Reactor burnup is tied to the amount of separative work indirectly, by means of the fuel requirements. If the required fuel mass is larger, more separative work is required. The length of the refueling interval affects it in a similar way. The combined effects of varying all three inputs shows that both the high and low estimate scenarios require less separative work than the baseline scenario. This suggests that the baseline scenario is poorly optimized with regards to separative work requirements. However, factors like neutronics and thermodynamics will place restrictions on what combinations of burnup and enrichment will be viable in real life applications. Consideration of these factors is beyond the scope of this analysis.

As one would expect, requiring spent fuel to spend a longer amount of time in interim storage results in increased storage capacity needs. This is evident from Figure 21c. One would also expect the total storage needs to be proportional to the amount of fuel consumed by the fleet. This can be seen indirectly in how reactor burnup affects the total storage needs. Higher burnup of the fuel in the reactor leads to lower overall fuel consumption, leading to less spent fuel being disposed of. One observes that the peak combined fuel storage needs are slightly higher for longer refueling intervals, while shorter refueling intervals produce a negligible change from the baseline. Overall, the effect of changed refueling intervals remains small. The combined effect of burnup, enrichment and refueling interval is dominated by the effect of the burnup.

For the amount of truckloads needed to remove spent fuel from the interim storage facilities, the amount

of fuel consumed plays a large role. Therefore, higher reactor burnup leads to fewer truckloads of spent fuel needing transportation yearly, as can be seen in Figure 21d. The storage time has no effect on the peak number of truckloads needed. A notable feature of Figure 21d is how the refueling interval affects the number of truckloads. The explanation behind this is that the amount of spent fuel being transported out in a given year is equal to the amount of fuel being loaded into the fleet $x + y$ years prior, where x is the refueling interval and y is the storage time. If the refueling interval is larger, more fuel must be loaded onto the ship at a time, resulting in more spent fuel being transported as well. Similarly, the transportation needs are lower for shorter refueling intervals. Similar to the storage needs, the truckload needs are dominated by the effects of varying the reactor burnup, with the length of the refueling interval playing a comparatively smaller role.

9 Conclusion

This study developed a simplified scenario model to estimate the fuel-cycle, maritime-network and port-infrastructure requirements associated with the deployment of an 800-vessel nuclear-powered maritime fleet between 2035 and 2060. The results should therefore be interpreted as first-order infrastructure estimates rather than as a detailed engineering design or forecast.

The main supply-chain bottlenecks identified for the 2060 fleet are HALEU enrichment, TRISO fuel fabrication and reactor-module production. By 2060, the baseline scenario requires approximately 240 tU of HALEU fuel per year, corresponding to about $3.0 \cdot 10^7$ TRISO pebbles. This is an order of magnitude above the TRISO production capacity and ~ 3 times larger than the HALEU production capacity assumed for the early deployment period. The enrichment requirement reaches approximately 5.7 MSWU/year by 2060, while the natural-uranium feed requirement reaches about 10,000 tU/year. In addition, the final fleet requires 1824 reactor modules of 25 MWe each. Although reactor manufacturing is not quantified as a capacity-constrained module in the present model, the number of required modules indicates that reactor production could become an additional bottleneck in a real deployment scenario.

The main infrastructure constraints are associated with port-based fuel storage and spent-fuel transport. The model shows that fresh-fuel storage rises during the main fleet-growth phase, while spent-fuel storage becomes dominant later because spent fuel is generated only after refuelling and must remain in interim storage before removal. By 2060, the global port network must manage approximately 3.4×10^5 m^3 of spent-fuel storage and 3.0×10^4 m^3 of fresh-fuel storage. The mature network contains 16 nuclear-capable ports, which distribute this burden geographically, although the demand remains concentrated at the highest-weighted service hubs. The largest ports, Algeciras and Shanghai, each require approximately 3.4×10^4 m^3 of combined fresh- and spent-fuel storage and must be prepared to handle about 400 spent-fuel truckloads per year. This indicates that port-based interim storage, cask handling and spent-fuel transport cadence are likely to be important constraints for a nuclear maritime system.

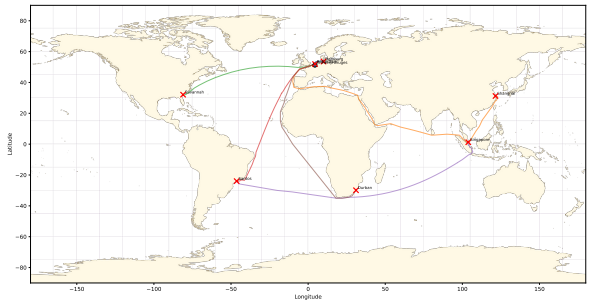
The sensitivity analysis shows that reactor-specific assumptions strongly affect the magnitude of the results. Higher burnup reduces the amount of fuel required, and therefore also reduces TRISO production, storage volume and spent-fuel transport requirements. Fuel enrichment mainly affects the separative-work requirement, while the refuelling interval influences both the timing and size of fuel loadings and spent-fuel outflows. If a maritime reactor could achieve burnup closer to that of advanced concepts such as the Xe-100, the fuel, storage and transport requirements could be reduced substantially, almost by a factor of two in the high-burnup sensitivity case. However, determining whether such burnup, enrichment and refuelling intervals are realistic for a compact maritime reactor would require neutronic, thermal-hydraulic and engineering analysis beyond the scope of this study.

The main limitations of the present model are the simplified treatment of ship operation, reactor design, route allocation, port capacity and regulation. The model does not perform detailed ship-design integration, reactor neutronics, port-engineering design, safety assessment, licensing analysis, cost optimisation or a quantitative reactor-manufacturing capacity assessment. It also assumes that the required reactor modules and fuel-cycle capacities can be expanded according to the baseline scenario. Future work could extend the model by introducing a reactor-supply module, explicit manufacturing and licensing timelines, more detailed port-capacity constraints, alternative geographical deployment scenarios, and economic or emissions comparisons with other low-carbon maritime fuels. Despite these limitations, the results show that nuclear maritime propulsion at the scale considered here would not only be a reactor or ship-design challenge, but a coupled infrastructure challenge involving fuel production, route networks, nuclear-capable ports, interim storage and spent-fuel transport.

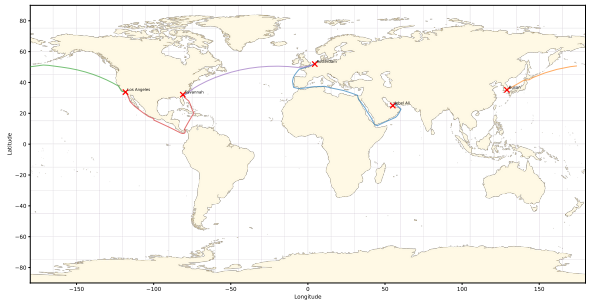
A Appendix: Ship-segment route maps for the 2060 maritime network

This appendix provides the ship-segment route maps generated by the maritime network model for the final year of the baseline scenario, 2060. The main text presents the aggregate active route network for all ship segments together, while the figures below separate the same mature network by vessel category. This makes it easier to identify which parts of the global route network are associated with bulk carriers, container ships, offshore vessels, oil tankers, chemical tankers, icebreakers, fishing vessels, cruise ships and passenger ferries.

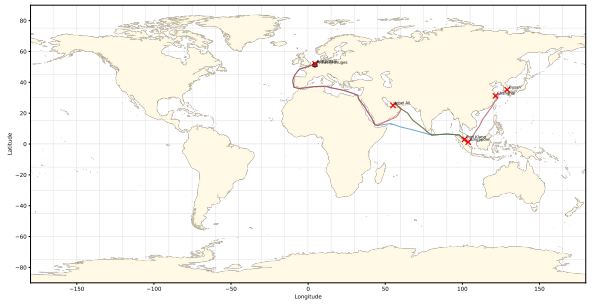
The maps are generated from the route-allocation model described in Section 5.2. For each ship segment, only the routes that are active in 2060 are shown. Ports marked with red crosses correspond to nuclear-capable ports that are active by 2060 according to the nuclear-port deployment assumptions introduced in the main text. Ordinary operating ports are shown as standard point markers. These figures therefore complement the aggregate 2060 route map by showing the spatial role of each ship segment in the mature maritime network.



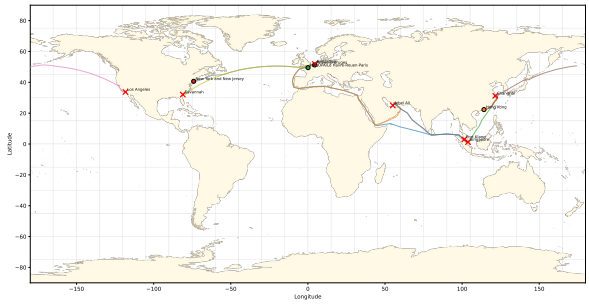
(a) Bulk carrier routes.



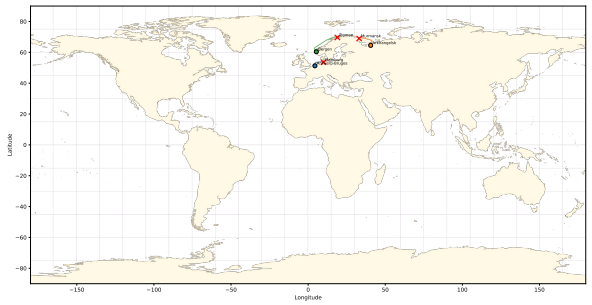
(b) Panamax container routes.



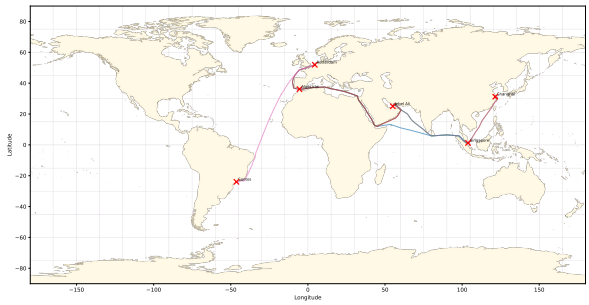
(c) Large container routes.



(d) Ultra-large container routes.

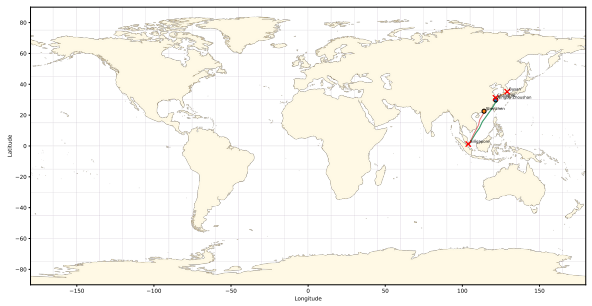


(e) Offshore vessel routes.

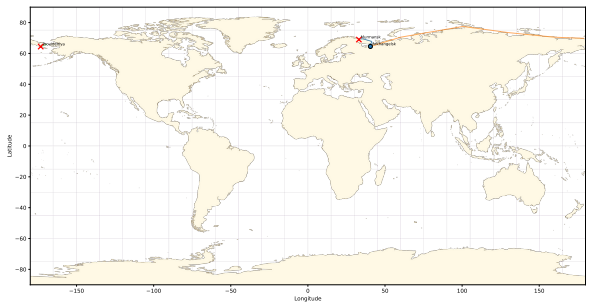


(f) Oil tanker routes.

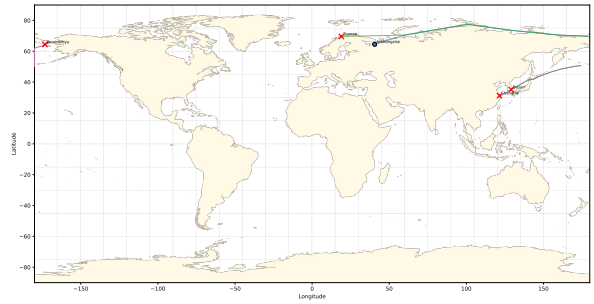
Figure 22



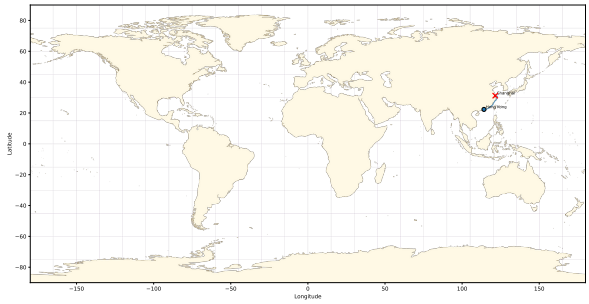
(a) Chemical tanker routes.



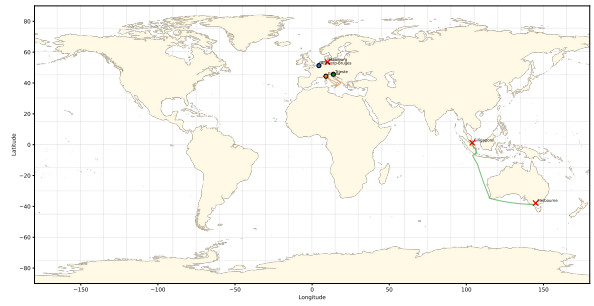
(b) Medium icebreaker routes.



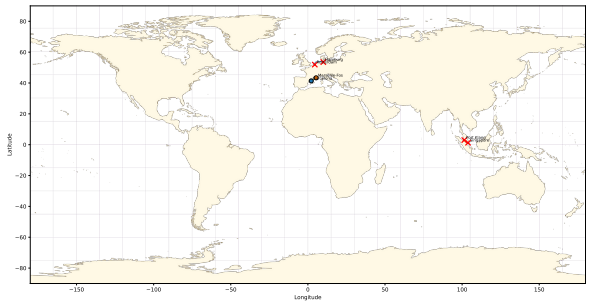
(c) Large icebreaker routes.



(d) Fishing-vessel routes.



(e) Large cruise routes.



(f) Passenger ferry routes.

Figure 23

B Appendix: Qualitative reactor-supply regionalisation

Table 20 provides a qualitative example of how a future reactor-supply module could be regionalised. It is included to illustrate one possible direction for extending the model beyond the current fuel-cycle and port-service calculations.

The regionalisation is based on the current Allseas offices, yards and logistics locations listed in Ref. [11]. These locations are not interpreted as existing nuclear manufacturing sites. Instead, they are used as qualitative indicators of where future reactor manufacturing, assembly, testing, maintenance or logistics-support capacity could potentially be developed if the maritime SMR deployment scenario were expanded into a full reactor-supply-chain model.

Table 20: Qualitative example of possible future reactor-supply regionalisation [11]. This table is not used as an input to the numerical model and has no effect on the results presented in Sections 6-8.

Period	Assumed supply region	Existing Allseas basis	Assumed role in the model
2030	Netherlands / Dutch industrial base	Engineering offices in Delft, Eindhoven and Enschede; logistics in Rotterdam; fabrication facilities in Heijningen and Hardinxveld-Giessendam	First dedicated reactor production, prototype qualification, early module assembly, testing and supply for the initial European deployment.
2038	Southeast Asia	Allseas office in Kuala Lumpur, close to the Singapore–Port Klang maritime corridor	Regional assembly, logistics and service-support capability for Asian and Southeast Asian deployment, reducing dependence on European supply for the fastest-growing route network.
2041	Middle East	Allseas office in Doha, close to the Jebel Ali route hub	Regional logistics and maintenance support for the Europe–Middle East–Asia corridor, especially for vessels operating through the Suez and Indian Ocean routes. In the model's country definition, the United Arab Emirates is associated with the SMR manufacturing instead of Qatar, as the port of Jebel Ali takes part in the port network.
2047	North America	Allseas USA office in Spring, Texas	Regional project office and possible future assembly, maintenance or supply-support hub for North American routes and transatlantic deployment.
2054	South America	Allseas offices in Rio de Janeiro	Additional regional support capacity for South Atlantic, routes.
2055	Oceania	Allseas offices in Perth	Additional regional support capacity for Australian as the fleet approaches its mature 2060 size.

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